

Minding the Meal Gap: Identifying Food Insecurity & SNAP Underutilization with the Food Bank of Northeast Georgia

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The Food Bank of Northeast Georgia serves a critical role in addressing Food Insecurity across 15 counties in Georgia. Despite its extensive efforts, Supplemental Nutrition Assistance Program (SNAP) participation remains underutilized. This project aims to identify areas of SNAP underutilization and explore factors contributing to these trends. By integrating datasets from *Map the Meal Gap*, the U.S. Census Bureau, the U.S. Department of Agriculture (USDA), and the U.S. Department of Transportation (DoT), we analyze patterns in Food Insecurity, demographic characteristics, and geographic accessibility. Our findings highlight key disparities such as wealth inequality and transportation barriers, and their impact on vulnerable and underrepresented populations.

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I. INTRODUCTION & BACKGROUND

Food Insecurity (FI) remains a significant challenge in the United States, particularly in rural and economically disadvantaged regions. Approximately 1,426,880 people in Georgia alone face hunger, including 461,720 children,

indicating that 1 in 8 individuals and 1 in 5 children are food insecure. [4] The Supplemental Nutritional Assistance Program (SNAP), formerly known as Food Stamps, is a federal initiative designed to provide assistance in purchasing food to low-income individuals and families. In Georgia, SNAP benefits are distributed via Electronic Benefit Transfer (EBT) cards, which recipients can use to purchase eligible food items at authorized retailers. These monthly benefits are eligible to Georgia residents that meet the following requirements: [9]

- Having a current bank balance (total value of savings and checking account amounts combined) under \$2,001 OR a current balance under \$3,001.
- Sharing your household with: a person or persons age 60 and over OR a person with a disability (child, parent, spouse, or yourself).

Despite the availability of SNAP, underutilization persists in various regions across Georgia. Speculative factors contributing to underutilization may include a lack of awareness about the program, perceived stigma associated with receiving assistance, and logistical barriers to application and access. However, these factors have not been conclusively explored, indicating a need for further research to identify the specific causes of SNAP underutilization in Georgia.

This project aims to identify regions, populations, and demographic trends within Georgia where SNAP is underutilized, with a focus on the service region of the Food Bank of Northeast Georgia. By analyzing data from Feeding America's *Map the Meal Gap*, [4] the U.S. Census Bureau, [12] the U.S. Department of Agriculture (USDA), [13] and the U.S. Department of Transportation (DoT), [14] we seek to uncover patterns of FI and SNAP participation. Our objectives include assessing FI trends, identifying demographic correlates of SNAP underutilization, evaluating geographic accessibility to food resources, and investigating socioeconomic disparities that may influence program participation. We aim to provide actionable insights to inform outreach strategies, policy

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advocacy, and program development, improving SNAP utilization.

A. The Food Bank of Northeast Georgia: History

The Food Bank of Northeast Georgia was founded in 1992 in Athens, Georgia with a mission to address Food Insecurity and end hunger in the region. Operating as a 501(c)3 non-profit organization, the food bank runs two main distribution hubs in Athens, Georgia and Clayton, Georgia. Through these hubs the Food Bank of Northeast Georgia serves a substantial portion of the state, covering fifteen counties: Banks, Barrow, Clarke, Elbert, Franklin, Habersham, Hart, Jackson, Madison, Oconee, Oglethorpe, Rabun, Stephens, Towns, and White County. [1] The organization works in partnership with local food suppliers and grocery stores in order to distribute food to human service organizations, educational institutions, and faith-based groups.

B. The Map the Meal Gap Initiative: History

In 2011, Feeding America, a non-profit organization in the fight against hunger, launched the *Map the Meal Gap* (MTMG) initiative in order to quantify Food Insecurity across all U.S. counties. By combining data from the U.S. Census Bureau, Bureau of Labor Statistics, and other federal sources, MTMG provides annual estimates of FI rates, demographic patterns, and the average cost of meals. [4]

For the Food Bank of Northeast Georgia, MTMG data has been a vital tool in identifying the primary demographics affected by Food Insecurity within its 15-county service region. This data has been instrumental in pinpointing factors that limit food assistance providers. A study published in April 2024 by *The Georgia Virtue* [8] highlighted several challenges faced by these providers, including limited infrastructure for storing and distributing food, shortages in resources and personnel, difficulties in increasing awareness and engagement with eligible clients, coordination issues due to tight resources and a lack of system-wide information, gaps in application assistance for programs like SNAP—offered by only about 20% of agencies—and a shortfall in nutrition education, with only 25% of providers offering such services. Despite these crucial new findings, there are limited analyses of the demographic and geographic constraints limiting client-side access to these resources. [8]

C. Intended Analyses

We structured our analysis to address three key questions:

- **Map the Meal Gap:** What are the FI trends across counties within the Food Bank of Northeast Georgia’s service region over time?
- **SNAP:** What are the demographics and trends amongst SNAP-enrolled populations?
- **Kid & Senior Hunger:** Where are vulnerable (child and senior) FI populations most at risk?

We first conducted a general overview observing FI trends in order to identify longitudinal patterns of FI within the service region and explore correlations with socioeconomic and demographic factors. Using MTMG data spanning the past decade, we analyzed changes in FI rates, meal costs, and their correlation with variables like median income and marriage status. [4] Spatial visualization through choropleth maps highlighted these trends across counties.

We observed trends across SNAP-enrolled populations by generating interactive choropleths with data sourced from the USDA Food Atlas [13] to uncover barriers to SNAP enrollment in regions with high FI but low participation. We analyzed enrollment patterns across different demographic groups and geographic regions, investigating the influence of more geographic factors like food source proximity, vehicle ownership, public transportation availability, and median income of neighboring Census Tracts and counties.

Lastly, our analysis of populations at risk incorporated data on low-access populations, particularly children and seniors living more than one mile from a food source, and explored how transportation availability, distance to food resources, and local funding contributed to FI in these groups. School district enrollment and per-student funding were also assessed to highlight counties requiring targeted interventions.

II. METHODS

A. Data Collection

Four primary data sources were utilized throughout this project: *Map the Meal Gap*, the USDA, the US Census Bureau, and the US News. Food insecurity demographics and meal costs were sourced from *Map the Meal Gap*, annual SNAP participation and food atlas data were sourced from the USDA, general population demographics were sourced from the U.S. Census Bureau, and school district data was sourced from the U.S. News. [4] [5] [12] [13] [14]

Geographical data for county-level analyses was sourced from the Georgia Geospatial information Office (GAGIO) [5]. For Census Tract geographies, we utilized Eric Celeste’s converted dataset of cartographic Census Tract data which had reformatted U.S. Census Bureau shapefiles into a more usable format. [1] [12] Transportation data was sourced from the DoT. [14]

B. Data Refinement

The data refinement process was critical to ensuring consistency, and usability across our diverse datasets. Raw data from sources like MTMG, [4] the USDA, [13] and the U.S. Census Bureau [12] were provided in various formats, including Comma-Separated Value (CSV) (.csv), Microsoft Excel (.xlsx, and JavaScript Object Notation (JSON) (.json). These files were standardized into uniform DataFrames using the Python library `pandas`, [7] [10] which enabled more seamless analyses using Python-based tools.

Some geographic data from the GAGIO [5] and from the U.S. Census Bureau [1] required additional conversion from JSON to GeoJSON formats using the Python library `geopandas` [6] for spatial mapping purposes to enable the code to correctly interpret and process geometries encoded within the data.

To narrow our scope, we first filtered national datasets by state to focus exclusively on data within Georgia. To address inconsistencies, column names were unified across datasets, standardizing terminology for key variables such as "county name" and resolving discrepancies like "Clayton County" versus "Clayton."

Missing data presented another challenge, with incomplete entries either imputed using county-level averages or removed when imputation was infeasible. For example, missing SNAP participation rates were estimated using adjacent years' data when possible, and missing median income data for specific Census Tracts was substituted with host county median income for mapping purposes. Spatial alignment was also prioritized, ensuring geographic data like tract-level FI rates matched county boundaries. Error identification and correction were integral to this process; scripts flagged mismatched column headers and duplicate entries for manual review.

Outliers, such as unusually high meal costs, were cross-validated with primary sources to maintain reliability. Finally, numeric variables were scaled or normalized, such as adjusting income levels for inflation and normalizing data to a 0–1 range for later correlation analyses.

C. Data Analysis

The analysis process for SNAP-enrolled populations began with plotting general data trends to establish a foundational understanding of the geographic and demographic context. Using the USDA Food Atlas [13] and MTMG [4] data, we visualized key metrics including FI rates, median income, and low-access demographics (e.g., children, seniors, SNAP-enrolled households) across Georgia counties and tracts.

Transportation metrics were subsequently integrated into the analysis to evaluate their role in exacerbating or mitigating Food Insecurity. Public transit routes, available as GeoJSON data, were overlaid onto FI maps to assess the accessibility of food sources and visualize how

well transportation infrastructure aligned with high-FI areas.

The next phase of analysis focused on examining the interaction between wealth disparity and FI. Wealth disparity was calculated as the absolute difference in median income between neighboring Census Tracts using adjacency data derived from spatial weights. These calculated disparities were mapped alongside FI rates to visually explore spatial correlations. Regions with high economic inequality were compared against their corresponding FI rates to identify patterns, such as the clustering of high-FI areas near wealthier tracts.

To validate these observations, statistical analyses were conducted. Correlation coefficients quantified the relationship between wealth disparity and Food Insecurity rates and regression modeling was performed to determine whether wealth disparity could predict FI rates. Finally, t-tests were conducted to compare FI rates in tracts with high and low wealth disparities. High-disparity and low-disparity tracts were identified based on percentile thresholds, and their FI rates were statistically compared, with results discussed in Section III C.

III. OBSERVATIONS & DISCUSSION

A. Trends Over Time: Financial Insecurity

Using data obtained from the US Census Bureau, [12] several different population factors were examined as they changed over time to identify potential indicators of SNAP participation and financial insecurity.

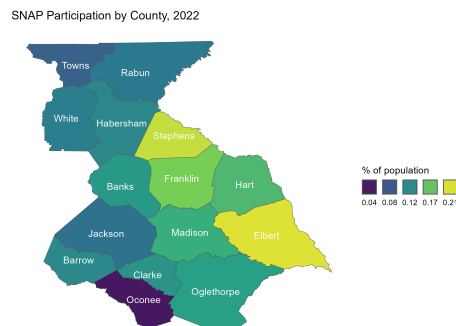


FIG. 1. Percent of the population enrolled in SNAP by county in 2022. [12]

Figure 1 represents SNAP participation across the service region in 2022. However, one aspect that is difficult to display in a choropleth is the decreasing trend in SNAP participation over the years. Although there was a minor spike in SNAP participation in 2020 due to the COVID19 pandemic, the increase was temporary and the successive years continue in a decreasing trend.

Figure 2 represents median individual income in 2022. It is important to note that this statistic includes all members of the populace, including those who cannot

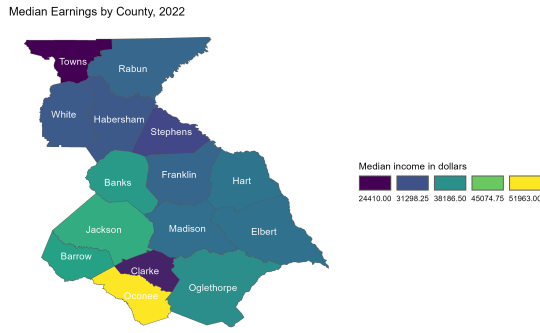


FIG. 2. Median individual income by county in 2022. [12]

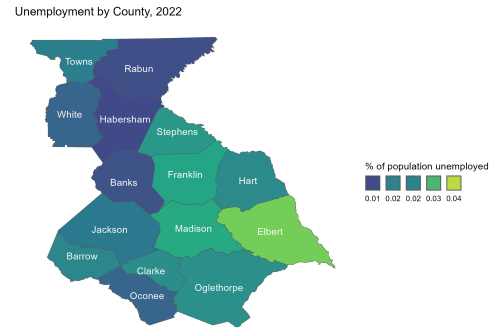


FIG. 4. Percent of the population unemployed by county in 2022. [12]

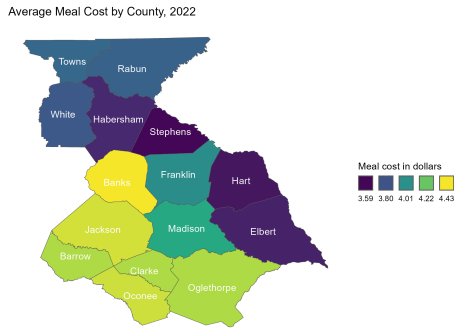


FIG. 3. Average cost per meal by county in 2022. [12]

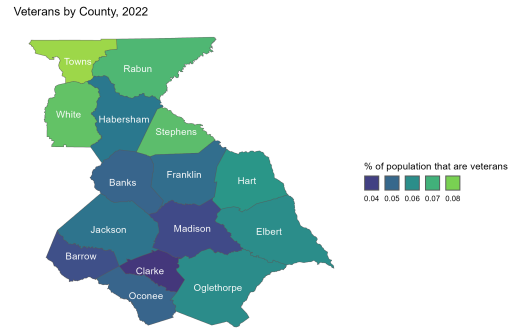


FIG. 5. Percent of the population that are veterans by county in 2022. [12]

work, are unemployed, or are retired. Unlike SNAP participation, median individual income has an upwards trend, with Oconee County having the highest median income in the entire region by a large margin. Clarke County stands as an interesting outlier, having a relatively low median income compared to its surrounding neighbors, but this disparity can be explained by taking note of the number of college students in Clarke County contrasted against the rest of the service region.

Figure 3 represents the average meal cost in 2022. Similarly to median individual income, meal costs have been increasing over time. However, instead of a gradual increase in meal costs over time, there is a sharp and significant increase in meal costs following the COVID19 pandemic - average meal costs nearly doubled in the three year period between 2020 and 2022.

Figure 4 represents the unemployment rate across the service region in 2022. Unemployment was found to have a positive correlation to SNAP participation, indicating that SNAP participation increases when unemployment increases and vice versa. For the service region, unemployment has been steadily decreasing over time with a minor increase during the COVID19 pandemic. Figure 5 represents the percentage of the population that are veterans in 2022. Similarly to unemployment, veteran populations positively correlate with SNAP participation and have also been steadily decreasing over time without any noticeable irregularities. Figure 6 represents the percent

of population that earn below the poverty line in 2022. Once again, the number of individuals under the poverty line is positively correlated with SNAP participation and also decreasing over time.

Figure 7 represents the percent of the population that are over 65 in 2022. Contrarily to what was expected, the elderly population was found to have a negative correlation with SNAP participation, meaning that SNAP participation was observed to decrease when the number of seniors increased. Figure 8 represents the percent of

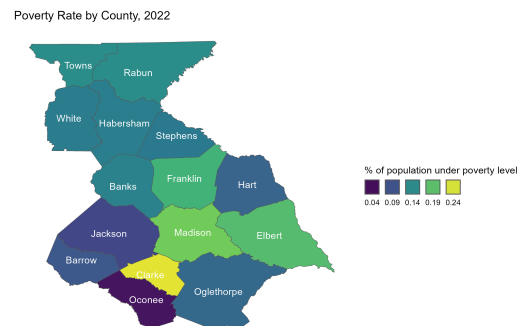


FIG. 6. Percent of the population under the poverty line by county in 2022. [12]

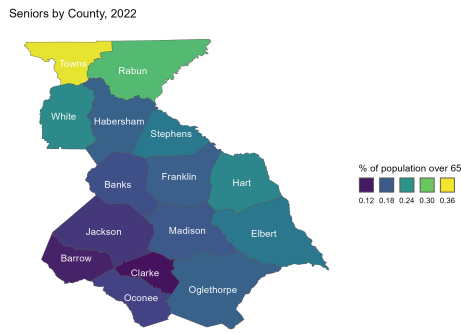


FIG. 7. Percent of the population that are over 65 by county in 2022. [12]

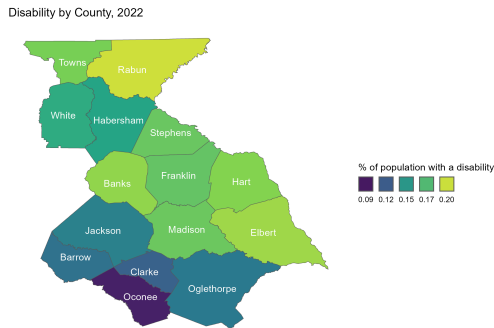


FIG. 8. Percent of the population with a disability by county in 2022. [12]

the population with a disability in 2022. Also unexpectedly, the percent of disabled individuals in the service region has a negative correlation with SNAP participation. Considering that both seniors and disabled individuals have fewer restrictions to obtaining SNAP eligibility, this is a strange phenomenon that potentially indicates an under-utilization of SNAP among these demographics. Figure 9 represents the percent of the population currently divorced in 2022. Like the two previous factors, divorcees have a negative correlation with SNAP partici-

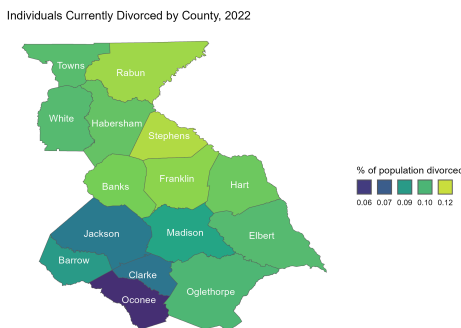


FIG. 9. Percent of the population currently divorced by county in 2022. [12]

pation, indicating potential under-utilization within this demographic as well.

B. County Demographics: 2018-2022 Trends

Utilizing data compiled from the Food Bank of North-east Georgia's Map the Meal Gap, we created demographic trend lines over the past five years. This line graph, shown in Figure 10 represents the trends in SNAP Participation by County through the years of 2018 to 2022. A noticeable decline in the participation rate was observed across several counties, a trend that continued into the following year of 2020 to 2021. The most striking shift occurred in 2020, when participation spiked unexpectedly. This surge can be largely attributed to the temporary expansion of SNAP benefits under federal COVID-19 relief efforts. Notably, Elbert County exhibited the highest participation rate, while Oconee County experienced the lowest.

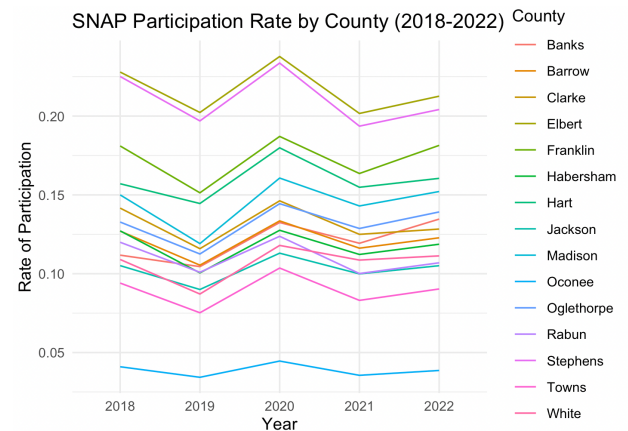


FIG. 10. SNAP Participation Rate by County 2018-2022 [4]

Several demographic and economic factors influence SNAP Participation. The bar graph, shown in Figure 11 represents the population density for each county in the year 2022. The population densities express the the average number of people living within a single square mile of land in that particular county. Counties with high population densities, such as Clarke County, typically indicate a more urbanized area. Contrastingly, counties with lower population densities, such as Oglethorpe County, indicate a more rural area.

Next, we graphed the food insecurity rate and its trend through the span of the year 2018 to 2022, shown in Figure 12. From 2019 to 2012, most counties experiences a declined in food insecurity rates. However, a noticeable spike occurred in 2022, with rates returning to or exceeding the levels in the year 2019. Among the counties, Franklin County exhibited the highest food insecurity rate, while Oconee County consistently had the lowest.

The unemployment rate for each county from the year

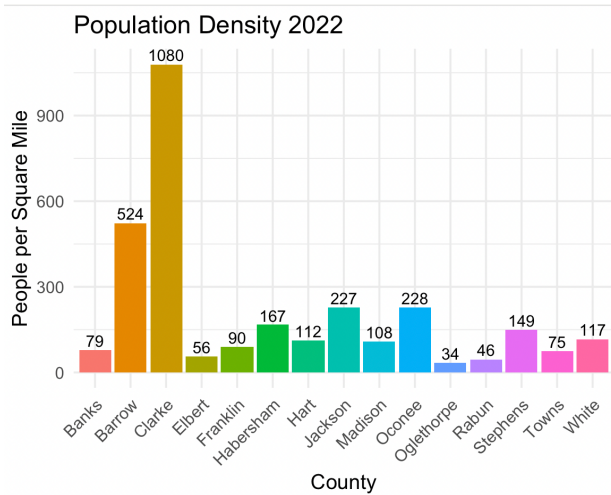


FIG. 11. Population Density by County 2022 ^[12]

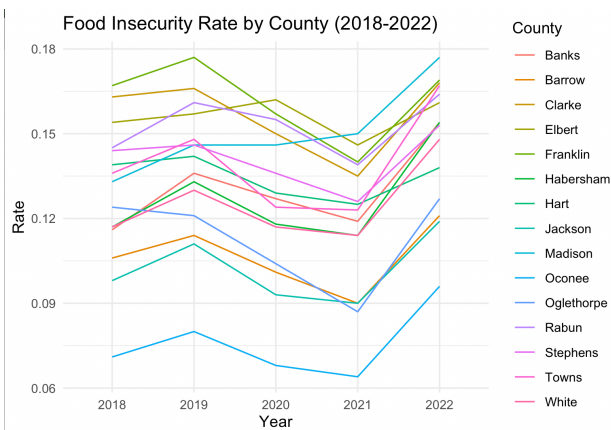


FIG. 12. Food Insecurity Rate by County 2018-2022 ^[4]

of 2018 to 2022 are illustrated in Figure ^[13]. The overall trend for all counties coincided with the economic atmosphere, surging during recession and declining during periods of recovery. This is represented in the huge spike in the unemployment rate in 2020, showing the impact of the COVID-19 pandemic. Towns County, Elbert County, and Stephens County experienced the highest unemployment rates, while Oconee County, Jackson County, and White County consistently had the lowest.

The gap in income levels between the fifteen counties from the span of 2019 to 2022 were expressed through the Figure ^[14]. From 2019 to 2022, the median income exhibited a consistent upward trajectory across all counties. Despite this overall increase, significant disparities persist between counties. Oconee County stands out with the highest median income while Elbert County reported the lowest median income.

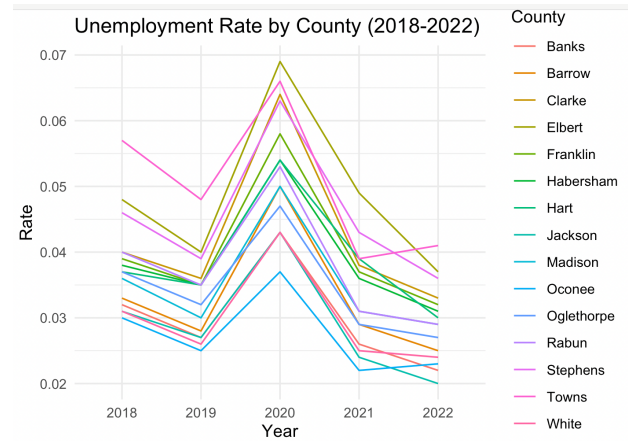


FIG. 13. Unemployment Rate by County 2018-2022 ^[4]

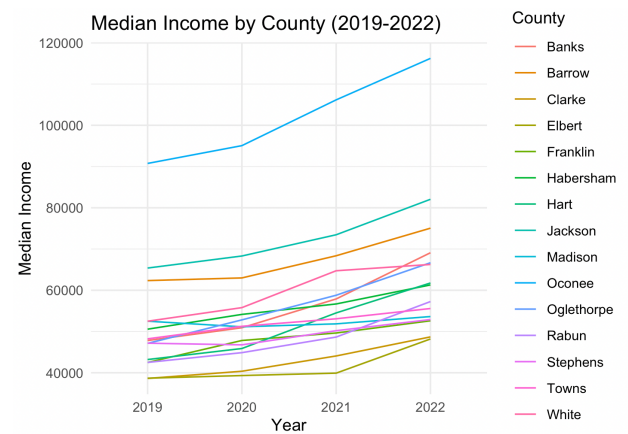


FIG. 14. Median Income by County 2019-2022 ^[4]

C. SNAP Participation: Limitations & Underutilization

We began by mapping FI rates by Census Tract across the Food Bank of Northeast Georgia's service region using 2022 MTMG ^[4] data. This choropleth, shown in Figure ^[15] represents the distribution of Food Insecurity within the region, emphasizing areas where FI is most acute and establishing a baseline understanding of where food assistance interventions may be prioritized.

Next, we mapped public transportation routes within the region, as shown in Figure ^[16]; we hypothesized that proximity to public transportation might alleviate FI by improving access to food resources. However, we observed an unexpected pattern: FI rates were higher in areas with public transportation routes, particularly in and around Athens-Clarke County, compared to more remote regions. One possible explanation is that areas with public transit may also have higher population densities and socio-economic disparities, potentially contributing to localized Food Insecurity.

A comparison of public transportation availability with

Food Insecurity Rate by Tract in Food Bank of Northeast Georgia's Service Region

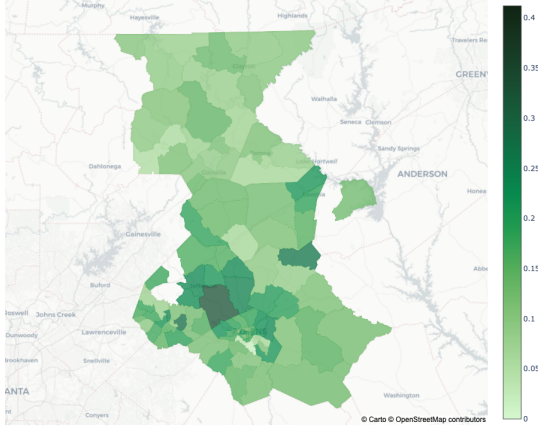


FIG. 15. Food Insecurity rate by tract across the Food Bank of Northeast Georgia's service region. Data sourced from *Map the Meal Gap*. [4]

low-access housing units (Figures [17] and [18]) revealed that Athens-Clarke County exhibits characteristically higher access, with fewer low-access housing units, compared to surrounding rural areas, suggesting that while transportation is a critical factor, other variables including affordability and awareness, could play equally significant roles. This pattern aligns with the observations in Section III A regarding SNAP participation trends, particularly the finding that SNAP enrollment has been declining despite increasing numbers of potentially eligible populations. The presence of public transportation, while important, may not be sufficient to overcome other barriers to food security and SNAP utilization, such as program awareness, stigma, and application accessibility.

Public Transportation Availability in Food Bank of Northeast Georgia's Service Region

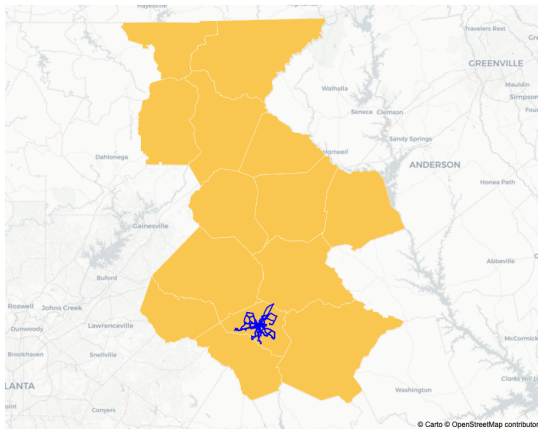


FIG. 16. Public transportation routes available within the Food Bank of Northeast Georgia's service region. Data sourced from the U.S. Department of Transportation. [14]

To explore whether wealth disparity between adja-

Share of Low-Access (1/2-Mile) SNAP-Enrolled Housing Units in Food Bank of Northeast Georgia's Service Region by Tract

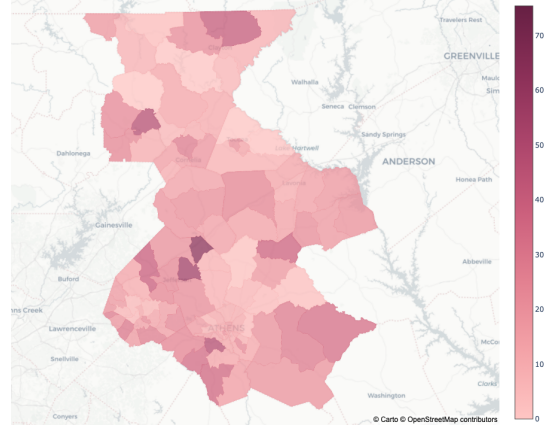


FIG. 17. Low-Access SNAP-Enrolled Housing Units ($\frac{1}{2}$ -mile from Food Source) that are across the Food Bank of Northeast Georgia's provider region by Census Tract. Data sourced from the USDA Food Atlas. [13]

Share of Low-Access (1-Mile) SNAP-Enrolled Housing Units in Food Bank of Northeast Georgia's Service Region by Tract

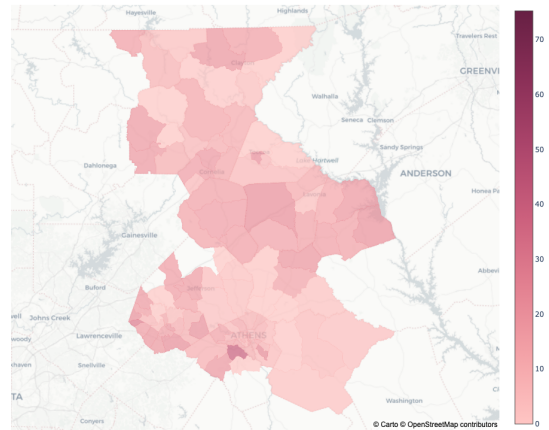


FIG. 18. Low-Access SNAP-Enrolled Housing Units (1-mile from Food Source) that are across the Food Bank of Northeast Georgia's provider region by Census Tract. Data sourced from the USDA Food Atlas. [13]

cent Census Tracts correlates with FI or low access, we mapped median incomes and identified tracts with the highest disparities (Figures [21] and [22]). Although our hypothesis was that higher wealth disparities would correlate with higher FI rates, statistical analysis revealed no significant correlation ($p > 0.05$). Similarly, low-access shares were only weakly and insignificantly correlated with wealth disparity, suggesting that wealth disparities may not directly influence FI but could interact with other factors, such as employment opportunities and housing costs, that create complex barriers to food access and SNAP participation. For instance, areas with high wealth disparities often feature:

- Higher housing costs that may force low-income

Share of Low-Access (1/2-Mile) Housing Units with No Vehicle in Food Bank of Northeast Georgia's Service Region by Tract

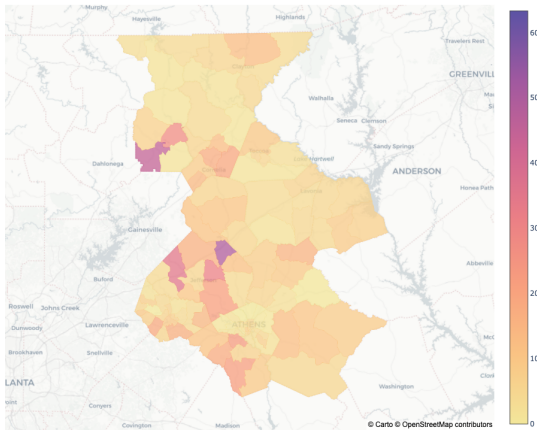


FIG. 19. A choropleth of Low-Access (1/2-Mile) Housing Units with no vehicle by Census Tract within the Food Bank of Northeast Georgia's service region. Data sourced from the USDA Food Atlas. [13]

Share of Low-Access (1-Mile) Housing Units with No Vehicle in Food Bank of Northeast Georgia's Service Region by Tract

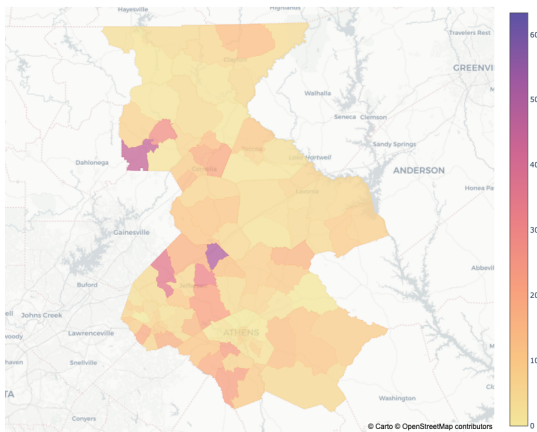


FIG. 20. A choropleth of Low-Access (1-Mile) Housing Units with no vehicle by Census Tract within the Food Bank of Northeast Georgia's service region. Data sourced from the USDA Food Atlas. [13]

residents to allocate more of their budget to rent, leaving less for food. [15]

- Fewer affordable grocery options, as retailers may cater to higher-income demographics. [3]
- Limited employment opportunities within walking distance, potentially increasing transportation costs. [15]
- Inadequate public transportation connecting low-income and high-income areas due to gentrification and socio-economic biases. [2]
- Social stigma around receiving assistance in mixed-income areas. [2]

Georgia Median Income (County & Tract) and Top Wealth Disparities

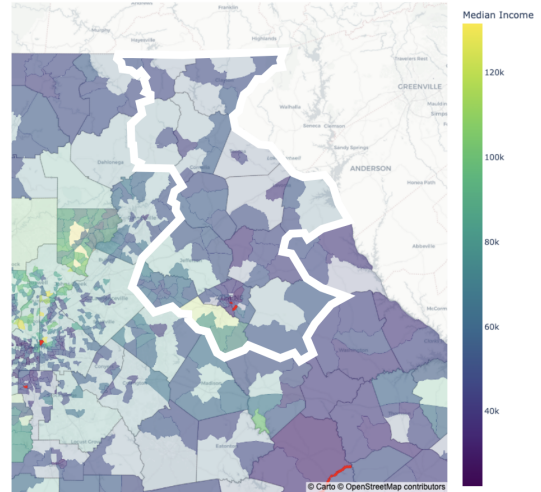


FIG. 21. Median income across the Food Bank of Northeast Georgia's service region. The red borders indicate neighboring Census Tracts within the region that display the highest wealth disparities (difference between median income between neighboring tracts). Data sourced from *Map the Meal Gap*. [4]

Georgia Median Income (County & Tract) and Top Wealth Disparities

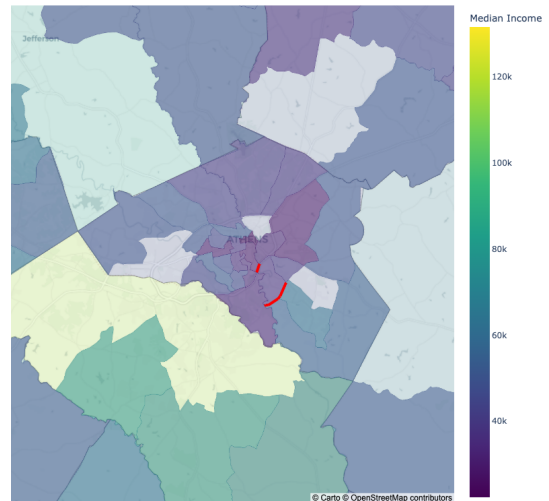


FIG. 22. Median income around Athens-Clarke County. The red borders indicate neighboring Census Tracts within the region that display the highest wealth disparities (difference between median income between neighboring tracts). Data sourced from the USDA Food Atlas and *Map the Meal Gap*. [13] [4]

Specific correlations between populations were thus identified in section ?? to supplement these weak correlations, suggesting a more nuanced interaction between demographic, geographic, and socioeconomic factors that, when combined, may point to populations at greater risk of FI and SNAP underutilization.

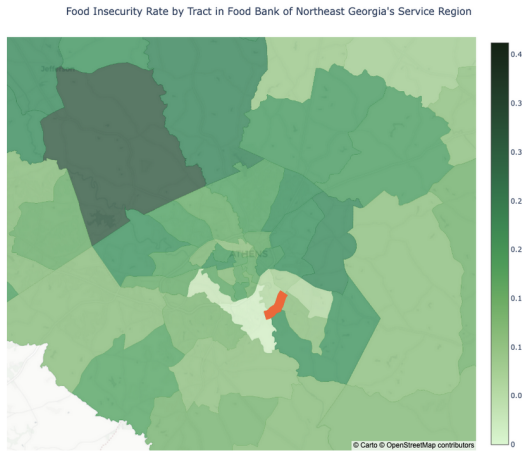


FIG. 23. Food Insecurity rate around Athens-Clarke County. The red border indicates neighboring Census Tracts within the region that display the highest wealth disparities (difference between median income between neighboring tracts). Data sourced from the USDA Food Atlas and *Map the Meal Gap*. [13] [4]

D. Populations at Risk: Child & Senior Hunger

Child and senior hunger is an immediate concern that serves as a focal point for the Food Bank of Northeast Georgia. [11] Beginning with the Food Insecurity rates over time, we analyzed the data over a 5 year period (2018-2022). In relation to child FI, we noticed a continuous decline in rates between 2018-2021. However, between 2021 and 2022, we revealed that the FI rate drastically began to increase (See Figures 24 and 25). Our hypothesis states that due to the Covid Pandemic, there were more resources and funding provided during the years of 2018-2021, thus a decrease among food insecure children. As the resources began stabilizing, nearing the decline of the pandemic, FI rates began to rise, as they had been prior to 2018. Breaking up this data by county, we reviewed the counties with the most consistent high child FI rates were Elbert, Clarke, Rabun, Madison, and Hart.

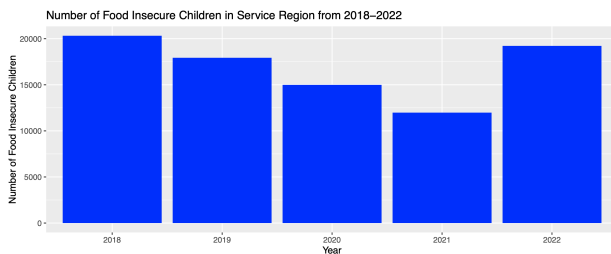


FIG. 24. Number of food insecure children within Food Bank of Northeast Georgia's service region during 5 year period (2018-2022).

Wanting to take a more detailed look into possible fac-

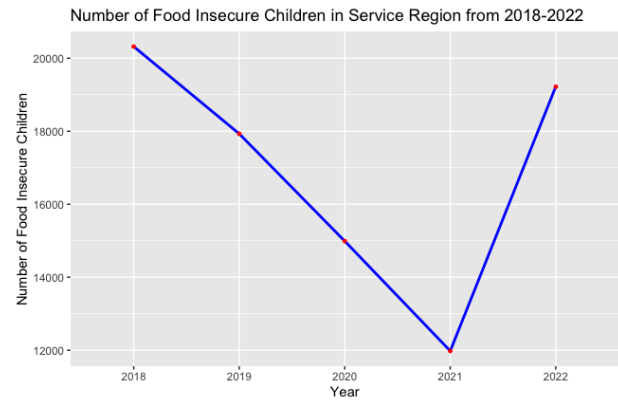


FIG. 25. Number of food insecure children within Food Bank of Northeast Georgia's service region over 5 year period (2018-2022).

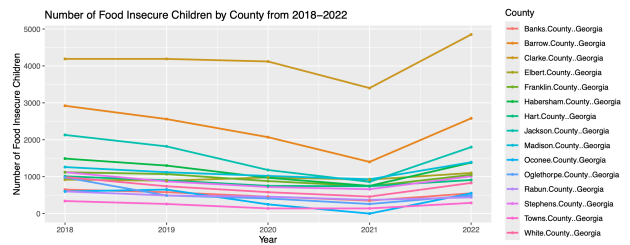


FIG. 26. Number of food insecure children within Food Bank of Northeast Georgia's service region by county from 2018-2022.

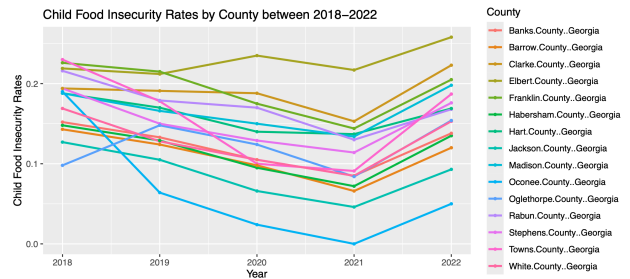


FIG. 27. Child Food Insecurity rates within Food Bank of Northeast Georgia's service region by county from 2018-2022.

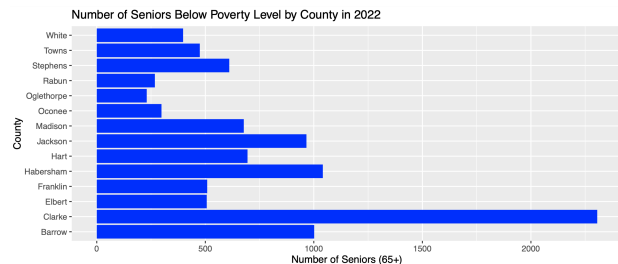


FIG. 28. Senior Food Insecurity within Food Bank of Northeast Georgia's service region by county in 2022.

tors and variables affecting these rates, we analyzed data

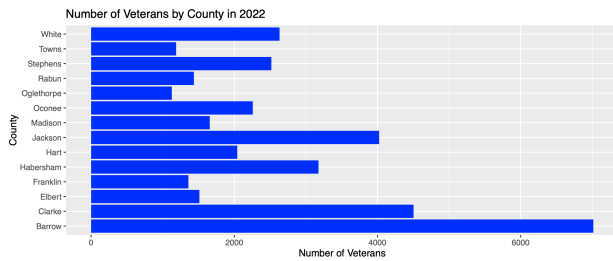


FIG. 29. Veteran status among county in Food Bank of Northeast Georgia's service region in 2022.

from the year 2022. Analyzing factors such as Food Insecurity rates, poverty rates, family units, veteran status, school district enrollment, and funding, we have discovered correlations among counties within the service region.

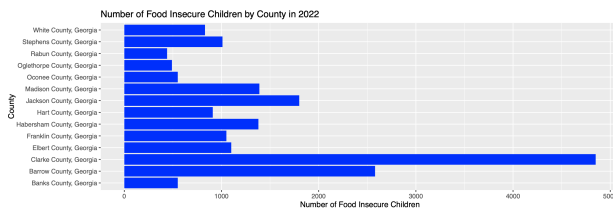


FIG. 30. Number of food insecure children within Food Bank of Northeast Georgia's service region by county during 2022.

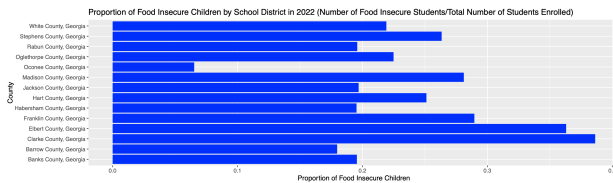


FIG. 31. Enrollment rates for school districts within the Food Bank of Northeast Georgia's service region.

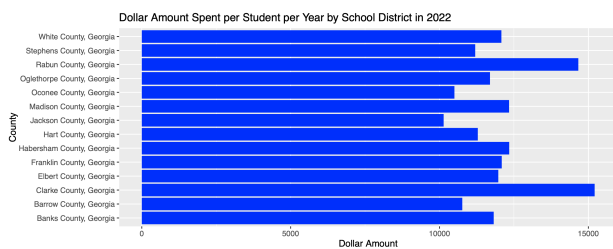


FIG. 32. School district funding and expense per student per year within Food Bank of Northeast Georgia's service region.

Studying child Food Insecurity, we first considered the actual number of food insecure children during this year. We identified the number of food insecure children within each county in 2022, (See Figure 30), as well as the total number of 19,220 food insecure children within the

service region. Researching the school districts in each county, we determined the Expense per Student per Year. This gave us insight into funding by county. We compared high school district enrollment and low funding and have discovered high child FI rates correlate with these variables within the following counties: Clarke, Madison, Barrow, and Jackson (See Figures 31 and 32). Diving into senior FI, we first analyzed the population of seniors below the poverty level. We separated this data by county, to show the counties with the highest poverty rates among seniors are Clarke, Habersham, and Barrow (See Figure 28). We wanted to determine if there was a correlation between FI rates and the veteran status in the counties (See FIG 16). We noticed the counties with the highest number of veterans were Barrow, Clarke, and Jackson. This coincides with the child and senior FI rates with these counties during 2022, with Clarke having the highest number of food insecure children and seniors. We then analyzed the number of children and seniors that reside more than 1 mile away from a supermarket (See Figures 34 and 33). We observed this data along with the number of housing units with no vehicle more than 1 mile from a supermarket, and number of housing units receiving SNAP benefits more than 1 mile from a supermarket. Furthermore, we determined the number of housing units receiving SNAP benefits and those with no vehicle (See Figures 35 and 36). These visuals display that the FI rates among children and seniors increase as the distance from a supermarket increases, shown through SNAP utilization and having no vehicle to access food.

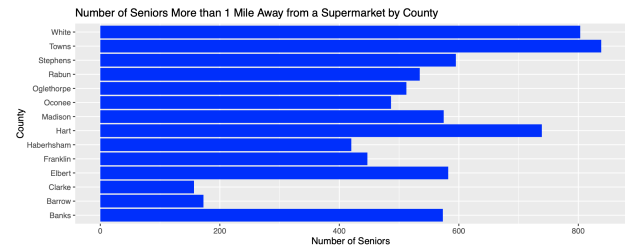


FIG. 33. Number of seniors located more than 1 mile from a supermarket within Food Bank of Northeast Georgia's service region by county in 2019.

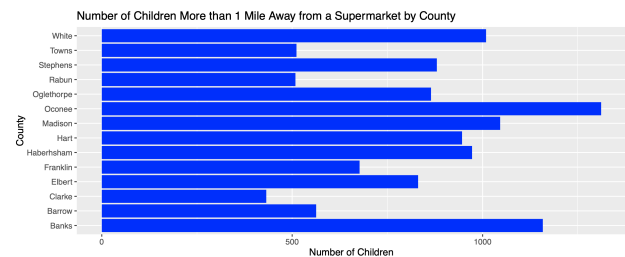


FIG. 34. Number of children located more than 1 mile from a supermarket within Food Bank of Northeast Georgia's service region by county in 2019.

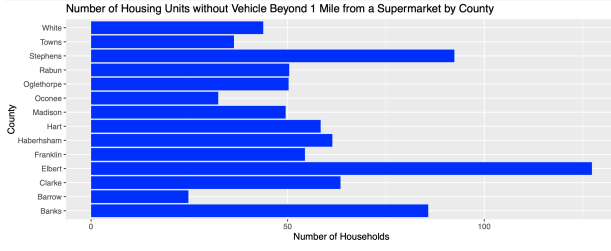


FIG. 35. Number of households located more than 1 mile from a supermarket with no vehicle within Food Bank of Northeast Georgia’s service region by county in 2019.

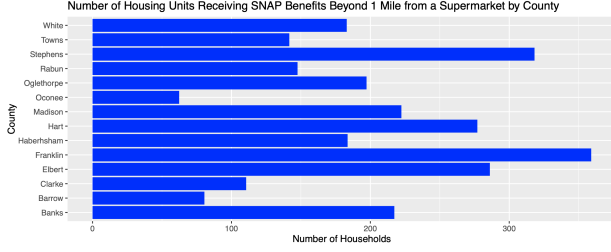


FIG. 36. Number of households located more than 1 mile from a supermarket receiving SNAP benefits within Food Bank of Northeast Georgia’s service region by county in 2019.

Our discoveries have relayed that variables such as transportation, accessibility, location, school district enrollment, and funding are likely high factors increasing FI among these at risk populations.

IV. CONCLUSION

Through our exploration of specific demographic, geographic, and socioeconomic trends across Georgia, we have nuanced our understanding of Food Insecurity and SNAP utilization across the Food Bank of Northeast Georgia’s service region.

First, we identified significant disparities in food access and SNAP participation that do not always align with traditional assumptions about urban versus rural Food Insecurity. While urban areas like Athens-Clarke County have better public transportation and physical access to food sources, they still experience high rates of FI, suggesting that accessibility extends beyond mere proximity to food sources.

Second, our analysis of demographic trends revealed concerning patterns of SNAP underutilization among several vulnerable populations. Despite increasing numbers of seniors, individuals with disabilities, and divorced individuals, SNAP participation has continued to decline, indicating potential gaps in program outreach and accessibility for these groups.

Third, our examination of child and senior FI highlighted the complex interplay between school district resources, geographic accessibility, and food security.

Counties with high school district enrollment but low per-student funding showed particularly high rates of child FI, suggesting that educational resource allocation may play a crucial role in addressing youth hunger.

A. Points of Action

The following points of action are informed by our findings and represent potential strategies to address identified challenges. These suggestions aim to serve as a starting point for consideration and is accompanied by examples to illustrate possible implementations, though these should be tailored to the specific needs and resources of each community.

Enhanced Targeted Outreach

- Develop specialized outreach programs for identified under-participating groups, particularly seniors, disabled individuals, and divorced populations.
- Create multilingual educational materials about SNAP benefits and application processes.
- Partner with community organizations that serve these populations to facilitate SNAP enrollment.

Transportation Solutions

- Implement mobile food pantry services in areas with high concentrations of low-access housing units.
- Develop volunteer transportation networks to connect seniors and disabled individuals with food sources.
- Coordinate with local transit authorities to optimize routes near food sources.

School-Based Initiatives

- Expand school breakfast and lunch programs in districts with high enrollment-to-funding ratios.
- Implement weekend food backpack programs in high-need areas.
- Create after-school meal programs in districts with high child Food Insecurity rates.

Food Access Improvements

- Partner with grocery stores to establish satellite locations in food deserts.
- Promote local farmers markets which accept SNAP benefits.
- Develop partnerships with organizations like *Misfits Market* or *Imperfect Produce* which retail produce that fail to meet grocery stores’ aesthetic criteria to provide low-cost healthy food options.

Administrative Barriers Reduction

- Simplify SNAP application processes, particularly for seniors and disabled individuals.
- Provide application assistance at food bank distribution sites.
- Create mobile enrollment units for rural areas.

B. Limitations

Our analysis faced several key limitations that should be considered when interpreting results:

Data Currency

- Some datasets, particularly transportation and food access metrics, were from 2019-2020.
- COVID-19 impact may have altered patterns since data collection.
- School district data may not reflect recent changes in enrollment or funding.

Geographic Resolution

- Census tract-level data sometimes masked micro-level disparities.
- Rural tract sizes made distance-based analyses less precise.
- County-level aggregation may have obscured important local variations.

Methodology Constraints

- Wealth disparity analysis considered only adjacent tracts, potentially missing broader patterns.
- Transportation analysis focused on public transit routes without accounting for frequency or reliability.
- Limited ability to track individual households over time.

Data Gaps

- Incomplete information about SNAP application attempts versus successful enrollments.
- Limited data on specific barriers to SNAP enrollment.
- Lack of qualitative data about individual experiences with Food Insecurity.

Temporal Limitations

- Analysis primarily focused on recent years, making long-term trend analysis difficult.

- Seasonal variations in Food Insecurity not fully captured.
- Limited ability to assess the impact of recent policy changes.

V. ACKNOWLEDGMENTS

The Food Bank Group I team thanks Dr. Jennifer Royal for her ongoing support and feedback throughout *Industrial Mathematics*. We thank the Food Bank of Northeast Georgia for providing *Mind the Meal Gap* data and for the opportunity to give back to the community.

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APPENDIX

Acronyms

CSV: Comma-Separated Value
DoT: U.S. Department of Transportation
FI: Food Insecurity
GAGIO: Georgia Geospatial Information Office
GeoJSON: Geographic JavaScript Object Notation
JSON: JavaScript Object Notation
MTMG: Map the Meal Gap
SNAP: Supplemental Nutrition Assistance Program
USDA: U.S. Department of Agriculture

Figures

All choropleths created with Python (Section [III C](#)) are available online at industrialmath-foodbank1.github.io. Please contact Anna Dmitrieff (annadmitrieff@uga.edu) and Dr. Jennifer Royal (jroyal@uga.edu) for any additional inquiries, error reports, or information requests.

Importing Libraries in Python

This section introduces the Python libraries used throughout the analysis and visualization process. The selected libraries include essential tools for data manipulation, geospatial analysis, statistical tests, and plotting. The included modules are:

- Pandas and NumPy for efficient data processing.
- Matplotlib and Plotly for visualization.
- Geopandas and Shapely for geospatial data handling.
- Scipy for statistical operations.

Additionally, a Mapbox access token is required for the interactive map visualizations. Make sure to follow the provided link to generate your token if you don't have one already.

```
1 import json
2 import plotly
3 import numpy as np
4 import pandas as pd
5 import plotly.io as pio
6 import geopandas as gpd
7 import matplotlib.pyplot as plt
8
9 from scipy.stats import ttest_ind
10 from libpysal.weights import Queen
11 from plotly import graph_objects as go
12 from plotly.subplots import make_subplots
13 from shapely.geometry import LineString, MultiLineString
14
15 # Mapbox Access Token
16 pio.templates.default = "plotly"
17 mapbox_access_token = "here is where you'd copy your token"
18 # For more information, visit https://docs.mapbox.com/help/getting-started/access-tokens/
```

Importing & Cleaning Data in Python

This section demonstrates how to:

- Load demographic, geographic, and specialized data like the Food Atlas and MTMG tract data.
- Standardize key identifiers (e.g., GEOIDs) to facilitate seamless merging of datasets.
- Filter and reformat columns to align data for further analysis.

```

1 # =====
2 # Census Demographic Data
3 census_df = pd.read_csv('path/to/csv/file')
4
5 # Rename 'CensusTract' to 'GEOID' and ensure it's a string
6 census_df = census_df.rename(columns={'CensusTract': 'GEOID'})
7 census_df['GEOID'] = census_df['GEOID'].astype(str)
8
9 # Pad GEOIDs to ensure they're all 11 characters (if not already)
10 census_df['GEOID'] = census_df['GEOID'].str.zfill(11)
11
12 # =====
13 # Census Geographic Data
14 # Load the Georgia census tracts GeoJSON into a GeoDataFrame
15 tracts_gdf = gpd.read_file('path/to/geojson/file')
16
17 # =====
18 # Georgia County Map Data
19 geo_georgia = gpd.read_file('path/to/geojson/file')
20
21 # Keep GeoJSON version
22 geo_georgia = json.loads(geo_georgia.to_json())
23
24 # For GeoDF version, ensure CRS is WGS84
25 geo_georgia_gdf = geo_georgia.to_crs(epsg=4326)
26
27 # =====
28 # Food Atlas Data
29 df = pd.read_excel('path/to/xlsx/file', sheet_name=2)
30 # Filter rows containing 'Georgia' in the 'State' column and create a
    new DataFrame
31 atlas_df = df.loc[df['State'].str.contains('georgia', case=False, na=
    False)].copy()
32
33 # Remove " County" suffix from the 'County' column
34 atlas_df['County'] = atlas_df['County'].str.replace(' County', '',
    regex=False)
35
36 # =====
37 # MTMG Tract Data
38 tract_df = pd.read_excel('path/to/xlsx/file', sheet_name=6)
39 # parses through 'State' column for strings containing 'Georgia'.
40 tract_df = tract_df[tract_df['state'].str.contains('ga', case=False,
    na=False)]
41
42 # =====
43 # MTMG County Data
44 county_df = pd.read_excel('/path/to/xlsx/file', sheet_name=1)
45 # Grabs Georgia data
46 county_df = county_df[county_df['State'].str.contains('ga', case=False
    , na=False)]
47
48 # formats county data as 'countyname' like other datasheets
49 county_df['County, State'] = county_df['County, State'].str.replace('

```

```

    County, Georgia', '', regex=False)
50 county_df.rename(columns={'Member 1': 'FoodBank'}, inplace=True)
51 county_df.rename(columns={'County, State': 'County'}, inplace=True)
52
53 # =====
54 # DOT Data
55 trans_geo = gpd.read_file('path/to/geojson/file')
56 trans_geo = trans_geo.to_crs(epsg=4326) # Ensure CRS is WGS84
57 trans_geo_json = json.loads(trans_geo.to_json())
58
59 # =====

```

Plotting Raw Data

Here, we explore the basics of visualizing data with Plotly's `Choroplethmapbox`.

```

1 # =====
2 # Plotting a single column of data
3 # =====
4 # Food Insecurity Rate by County
5 FIbyCounty = go.Figure(
6     go.Choroplethmapbox(
7         geojson=geo_georgia, # The GeoJSON
8         featureidkey='properties.name', # The key in
9         locations=county_df["County"], # The tract
10         z=county_df["Overall Food Insecurity Rate"], #
11         zauto=True, # Disable
12         colorscale="algae",
13         showscale=True, # Show color
14         marker_opacity=0.75, # Opacity of
15         marker_line_width=0 # No border
16     )
17 )
18
19 FIbyCounty.update_layout(
20     mapbox_accesstoken=mapbox_access_token,
21     mapbox_style="carto-positron",
22     mapbox_zoom=6,
23     mapbox_center={"lat": 32.5, "lon": -83.5},
24     title_x=0.5,
25     title_y=0.95,
26     title_text="Food Insecurity Rate by County",
27     margin=dict(
28         l=0,

```

```

29         r=0,
30         b=0,
31         t=90,
32         pad=4,
33         autoexpand=True
34     ),
35     width=900,
36     height=800
37 )
38
39 # Show the figure
40 FIbyCounty.write_html('path/to/save/file.html')
41 FIbyCounty.show()
42
43 # =====
44 # Making a custom color category
45 # =====
46
47 # Food Bank Providers by County
48 foodbank_colors = {
49     "America's Second Harvest of Coastal Georgia, Inc.": "#f94144", #
50         Imperial Red
51     "Second Harvest of South Georgia, Inc": "#90be6d", #
52         Pistachio
53     "Middle Georgia Community Food Bank": "#577590", #
54         Payne's Gray
55     "Food Bank of Northeast Georgia": "#f9c74f", #
56         Saffron
57     "Atlanta Community Food Bank": "#4d908e", #
58         Dark Cyan
59     "Golden Harvest Food Bank": "#f3722c", #
60         Orange Crayola
61     "Feeding the Valley Food Bank": "#43aa8b", #
62         Zomp
63     "Chattanooga Area Food Bank": "#f8961e" #
64         Carrot Orange
65 }
66
67 # Initialize the figure
68 foodbankscnty = go.Figure()
69
70 # For each food bank, add a trace
71 for fb_name, color in foodbank_colors.items():
72     # Filter the DataFrame for counties served by this food bank
73     df_fb = county_df[county_df['FoodBank'] == fb_name]
74
75     # Add the choropleth trace
76     foodbankscnty.add_trace(go.Choroplethmapbox(
77         geojson=geo_georgia,
78         featureidkey="properties.name",
79         locations=df_fb["County"], # The counties for this food
80         bank
81         z=[1]*len(df_fb), # Dummy z value

```



```

73         colorscale=[[0, color], [1, color]], # Single-color
           colorscale
74         marker_line_width=0.5,
75         marker_line_color='white',
76         showscale=False,
77         name=fb_name,
78         showlegend=True,
79         hoverinfo='location', # Display county name on hover
80     ))
81
82 # Update the layout
83 foodbankscnty.update_layout(
84     mapbox_accesstoken=mapbox_access_token,
85     mapbox_style="carto-positron",
86     mapbox_zoom=6,
87     mapbox_center={"lat": 32.5, "lon": -83.5},
88     title_x=0.5,
89     title_y=0.95,
90     title_text="Food Banks Serving Counties in Georgia",
91     margin=dict(
92         l=0,
93         r=0,
94         b=0,
95         t=90,
96         pad=4,
97         autoexpand=True
98     ),
99     width=900,
100    height=800,
101    legend=dict(
102        title="Food Banks",
103        itemsizing='constant',
104        traceorder='normal'
105    )
106 )
107
108 # Show the figure
109 foodbankscnty.write_html("save/path/for/file.html")
110 foodbankscnty.show()
111
112
113 # =====
114 # Plotting multiple categories (Low Access at 1, 10, and 20 miles) on
   the same choropleth
115 # =====
116
117 # Creating a new column/category for distance flags in order to plot
   them on the same choropleth:
118
119 atlas_df['distance_category'] = np.select(
120     [atlas_df['LATracts1'] == 1, atlas_df['LATracts10'] == 1, atlas_df
       ['LATracts20'] == 1],
121     [1, 10, 20],
122     default=0

```

```

123 )
124
125 LAtracts = go.Figure(
126     go.Choroplethmapbox(
127         geojson=geo_tracts,           # The GeoJSON
128         featureidkey='properties.GEOID', # The key in
129         locations=atlas_df["CensusTract"], # The tract
130         z=atlas_df["distance_category"], # Use the new
131         zauto=False,                  # Disable
132         zmin=1,                       # Set minimum
133         zmax=20,                      # Set maximum
134         colorscale=[                  # Custom
135             [0, "lightskyblue"],      # 1 mile (light green)
136             [0.5, "cornflowerblue"], # 10 miles (medium green)
137             [1, "midnightblue"],     # 20 miles (dark green)
138         ],
139         showscale=True,               # Show color
140         marker_opacity=0.75,          # Opacity of
141         marker_line_width=0,          # No border
142         on the tracts
143     )
144
145 LAtracts.update_layout(
146     mapbox_accesstoken=mapbox_access_token,
147     mapbox_style="carto-positron",
148     mapbox_zoom=6,
149     mapbox_center={"lat": 32.5, "lon": -83.5},
150     title_x=0.5,
151     title_y=0.95,
152     title_text="Low-Access Tracts at 1, 10, and 20 Miles from a Food
153         Source",
154     margin=dict(
155         l=0,
156         r=0,
157         b=0,
158         t=90,
159         pad=4,
160         autoexpand=True
161     ),
162     width=900,
163     height=800
164 )

```

```

165 # Show the figure
166 LAtracts.write_html("save/path/for/file.html")
167 LAtracts.show()

```

Plotting Combined Data

Combining datasets allows us to examine overlapping phenomena, such as public transportation availability within specific food bank service regions. This section covers:

- Filtering datasets to isolate relevant subsets, like counties within the Food Bank of Northeast Georgia's service area.
- Visualizing combined data on a single map, with layers representing different attributes (e.g., transit routes and food insecurity rates).

```

1
2 # =====
3 # Filtering for *only* counties within the Food Bank of Northeast
  Georgia's service region
4 fb_name = "Food Bank of Northeast Georgia"
5 # =====
6
7 # Ensure that the food bank exists in the dataset
8 if fb_name not in county_df['FoodBank'].unique():
9     raise ValueError(f"{fb_name} not found in the 'FoodBank' column of
      county_df.")
10
11 # Filter the DataFrame for the specific food bank
12 fb_counties = county_df[county_df['FoodBank'] == fb_name]['County'].
  tolist()
13
14 # =====
15 # Procedure if Food Bank provider is not available in dataset
16 # =====
17
18 # 1. Define the list of counties in the service region
19 list_of_counties_in_foodbank_service_region = [
20     'Banks', 'Barrow', 'Clarke', 'Franklin', 'Habersham',
21     'Hart', 'Jackson', 'Madison', 'Oconee', 'Oglethorpe',
22     'Rabun', 'Stephens', 'Townsend', 'White'
23 ]
24
25 # 2. Standardize county names in the DataFrame
26 atlas_df['County'] = atlas_df['County'].str.strip().str.title()
27
28
29 # 3. Standardize county names in the list (optional)
30 list_of_counties_in_foodbank_service_region = [
31     county.strip().title() for county in
      list_of_counties_in_foodbank_service_region

```

```

32 ]
33
34 # 4. Filter the DataFrame for the specified counties
35 filtered_df = atlas_df[
36     atlas_df['County'].isin(
37         list_of_counties_in_foodbank_service_region)
38 ].copy()
39
40 # 5. Select and rename relevant columns
41 filtered_tracts_df = filtered_df[['County', 'CensusTract']].copy()
42
43 # 6. Reset the index
44 filtered_tracts_df.reset_index(drop=True, inplace=True)
45
46 # 7. Verify the results
47 print(filtered_tracts_df.head())
48 print(f"Total number of tracts in FBNGA's service region: {
49     filtered_tracts_df.shape[0]}")
50
51 # =====
52 # Filtering for *only* tracts within the Food Bank of Northeast
53 # Georgia's service region
54 # =====
55
56 # Standardize county names in atlas_df
57 atlas_df['County'] = atlas_df['County'].str.strip().str.title()
58
59 # Standardize county names in filtered_tracts_df (if not already
60 # standardized)
61 filtered_tracts_df['County'] = filtered_tracts_df['County'].str.strip(
62     ).str.title()
63
64 # Extract the list of CensusTract IDs from filtered_tracts_df
65 service_region_tracts = filtered_tracts_df['CensusTract'].unique().
66     tolist()
67
68 # Filter atlas_df to include only tracts in service_region_tracts
69 filtered_atlas_df = atlas_df[atlas_df['CensusTract'].isin(
70     service_region_tracts)].copy()
71
72 print(f"Number of tracts in filtered_atlas_df: {filtered_atlas_df.
73     shape[0]}")
74
75 display(filtered_atlas_df.head())
76
77 filtered_tracts_list = filtered_tracts_df['CensusTract'].tolist()
78 filtered_tracts_list = [str(x) for x in filtered_tracts_list]
79
80 # for geo_tracts:
81 # Filter the GeoJSON features to include only the tracts in
82 # service_region_tracts
83 filtered_geojson = {
84     "type": "FeatureCollection",
85     "features": [

```

```

77         feature for feature in geo_tracts['features']
78         if feature['properties']['GEOID'] in filtered_tracts_list
79     ]
80 }
81
82 print(f"Number of tracts in filtered_geojson: {len(filtered_geojson['
    features'])}")
83
84 # Mapping the results (with the transportation routes as a trace)
85 # Initialize the figure
86 foodbank_scnty = go.Figure()
87
88 # Add the choropleth trace for Food Bank of Northeast Georgia
89 foodbank_scnty.add_trace(go.Choroplethmapbox(
90     geojson=geo_georgia,
91     featureidkey="properties.name", # Ensure this matches the county
        name property in geo_georgia
92     locations=fb_counties,          # The counties for this food bank
93     z=[1]*len(fb_counties),        # Dummy z value since we're using
        a single color
94     colorscale=[[0, "#f9c74f"], [1, "#f9c74f"]], # Saffron color for
        the food bank
95     marker_line_width=0.5,
96     marker_line_color='white',
97     showscale=False,
98     name=fb_name,
99     hoverinfo='location',          # Display county name on hover
100 ))
101
102 # Add the transit routes overlay
103 # Extract features from trans_geo
104 trans_features = trans_geo_json['features']
105
106 # Prepare lists to hold coordinates
107 trans_lons = []
108 trans_lats = []
109
110 for feature in trans_features:
111     geometry = feature['geometry']
112     if geometry['type'] == 'LineString':
113         # Extract only longitude and latitude
114         lons, lats = zip(*(coord[:2] for coord in geometry['
            coordinates'])))
115         trans_lons.append(lons)
116         trans_lats.append(lats)
117     elif geometry['type'] == 'MultiLineString':
118         for line in geometry['coordinates']:
119             # Extract only longitude and latitude for each line
120             lons, lats = zip(*(coord[:2] for coord in line))
121             trans_lons.append(lons)
122             trans_lats.append(lats)
123
124 # Add transit routes as lines
125 for lon, lat in zip(trans_lons, trans_lats):

```



```

126     foodbank_scnty.add_trace(go.Scattermapbox(
127         lon=lon,
128         lat=lat,
129         mode='lines',
130         line=dict(width=2, color='blue'),
131         hoverinfo='none',
132         name='Public Transportation',
133         showlegend=False
134     ))
135
136 # Update the layout
137 foodbank_scnty.update_layout(
138     mapbox_accesstoken=mapbox_access_token,
139     mapbox_style="carto-positron",
140     mapbox_zoom=8, # Adjusted zoom for better visibility of service
141     region
142     mapbox_center={"lat": 34.35, "lon": -83.35}, # Adjust center to
143     focus on Northeast GA
144     title_x=0.5,
145     title_y=0.95,
146     title_text="Public Transportation Availability in Food Bank of
147     Northeast Georgia's Service Region",
148     margin=dict(
149         l=0,
150         r=0,
151         b=0,
152         t=90,
153         pad=4,
154         autoexpand=True
155     ),
156     width=900,
157     height=800,
158     legend=dict(
159         title="Legend",
160         itemsizing='constant',
161         traceorder='normal'
162     )
163 )
164
165 # Show the figure
166 foodbank_scnty.show()

```

Plotting Modified Data & Generating Analyses

This section introduces more advanced analyses, focusing on wealth disparities and their potential correlation with food insecurity:

- Spatial adjacency weights are computed to identify neighboring tracts.
- Wealth disparity between neighbors is quantified based on shared geographic boundaries.

- The analysis examines whether higher wealth disparity correlates with food insecurity, using statistical metrics like Pearson and Spearman correlations and t-tests.

Finally, the results are visualized with overlays highlighting the areas of greatest disparity.

```

1 # =====
2 # filtering data
3 # Ensure the CRS is set to a projected system suitable for spatial
  operations
4 tracts_gdf = tracts_gdf.to_crs(epsg=3857) # Web Mercator projection
5
6 # Ensure tract IDs are strings and padded to 11 characters (if
  necessary)
7 tract_df['tractid'] = tract_df['tractid'].astype(str).str.zfill(11)
8 tracts_gdf['GEOID'] = tracts_gdf['GEOID'].astype(str).str.zfill(11)
9
10 # Merge on 'GEOID' and 'tractid'
11 merged_gdf = tracts_gdf.merge(tract_df[['tractid', 'median_income']],
    left_on='GEOID', right_on='tractid', how='left')
12
13 # checking your dataset routinely is always a good idea
14 display(merged_gdf.head())
15
16 # Identify non-numeric entries in 'median_income'
17 non_numeric_entries = merged_gdf[~merged_gdf['median_income'].apply(
    lambda x: str(x).replace('.', '', 1).isdigit())]
18
19 print("Non-numeric entries in 'median_income':")
20 print(non_numeric_entries[['GEOID', 'median_income']])
21
22 # Replace non-numeric entries with NaN
23 merged_gdf['median_income'] = pd.to_numeric(merged_gdf['median_income']
    ], errors='coerce')
24
25 # Drop rows with NaN in 'median_income'
26 merged_gdf = merged_gdf.dropna(subset=['median_income'])
27
28 # Then convert to integer
29 merged_gdf['median_income'] = merged_gdf['median_income'].astype(int)
30 # =====
31 # Compute spatial weights (adjacency)
32 weights = Queen.from_dataframe(merged_gdf)
33
34 # Get the neighbors dictionary
35 neighbors_dict = weights.neighbors
36 # =====
37
38 # calculating wealth disparities
39
40 # Prepare a set to store tract pairs and a list for disparities
41 seen_pairs = set()
42 disparities = []
43

```

```

44 for tract_index, tract_row in merged_gdf.iterrows():
45     tract_id = tract_row['GEOID']
46     tract_income = tract_row['median_income']
47     tract_geometry = tract_row['geometry']
48
49     # Get indices of neighboring tracts
50     neighbor_indices = neighbors_dict.get(tract_index, [])
51
52     # Iterate over neighbors
53     for neighbor_index in neighbor_indices:
54         neighbor_row = merged_gdf.iloc[neighbor_index]
55         neighbor_id = neighbor_row['GEOID']
56         neighbor_income = neighbor_row['median_income']
57         neighbor_geometry = neighbor_row['geometry']
58
59         # Create a sorted tuple of tract IDs to avoid duplicates
60         tract_pair = tuple(sorted([tract_id, neighbor_id]))
61
62         if tract_pair not in seen_pairs:
63             seen_pairs.add(tract_pair)
64
65             # Calculate income disparity
66             disparity = abs(tract_income - neighbor_income)
67
68             # Calculate shared boundary by intersecting full
69             geometries
70             shared_boundary = tract_geometry.intersection(
71                 neighbor_geometry)
72
73             # Ensure shared_boundary is a LineString or
74             MultiLineString
75             if shared_boundary.is_empty:
76                 # If no intersection, skip this pair
77                 continue
78             elif isinstance(shared_boundary, (LineString,
79                 MultiLineString)):
80                 disparities.append({
81                     'tract_id': tract_id,
82                     'neighbor_id': neighbor_id,
83                     'tract_income': tract_income,
84                     'neighbor_income': neighbor_income,
85                     'disparity': disparity,
86                     'geometry': shared_boundary
87                 })
88             else:
89                 # Handle cases where the intersection is a Point or
90                 other geometry
91                 # For adjacent tracts, it should primarily be
92                 LineString or MultiLineString
93                 continue
94
95 # Create the disparities DataFrame as before
96 disparities_df = pd.DataFrame(disparities)
97

```

```

92 # Sort by disparity in descending order
93 disparities_df = disparities_df.sort_values(by='disparity', ascending=
    False)
94
95 # Select top N disparities (e.g., top 20)
96 top_n = 60
97 top_disparities = disparities_df.head(top_n).reset_index(drop=True)
98
99 # Display the top disparities
100 display(top_disparities[['tract_id', 'neighbor_id', 'tract_income', '
    neighbor_income', 'disparity', 'geometry']])
101
102 # Convert top_disparities to a GeoDataFrame
103 top_disparities_gdf = gpd.GeoDataFrame(top_disparities, geometry='
    geometry', crs=merged_gdf.crs)
104
105 # Verify the GeoDataFrame
106 print(top_disparities_gdf.head())
107
108 # Check for invalid or empty geometries
109 invalid_geometries = top_disparities_gdf[top_disparities_gdf['geometry
    '].is_empty]
110 print(f"Number of disparities with empty geometries: {
    invalid_geometries.shape[0]}")
111
112 # Convert top_disparities_gdf to WGS84 CRS for Plotly
113 top_disparities_gdf_wgs84 = top_disparities_gdf.to_crs(epsg=4326)
114
115 # Convert merged_gdf to GeoJSON
116 merged_gdf_wgs84 = merged_gdf.to_crs(epsg=4326) # Convert to WGS84
    for Plotly
117 merged_json = json.loads(merged_gdf_wgs84.to_json())
118
119 # =====
120 # plotting
121
122 # Step 1: Create the choropleth map
123 fig = px.choropleth_mapbox(
124     merged_gdf_wgs84,
125     geojson=merged_json,
126     locations='GEOID',
127     featureidkey='properties.GEOID',
128     color='median_income',
129     color_continuous_scale="Viridis",
130     mapbox_style="carto-positron",
131     zoom=6,
132     center={"lat": 32.9, "lon": -83.5},
133     opacity=0.5,
134     labels={'median_income': 'Median Income'})
135
136
137 # Step 2: Add the top disparities as lines with converted coordinates
138 for idx, row in top_disparities_gdf_wgs84.iterrows():
139     geometry = row['geometry']

```

```

140     if geometry is not None and not geometry.is_empty:
141         if geometry.type == 'MultiLineString':
142             for line in geometry.geoms:
143                 x, y = line.xy
144                 lon = list(x) # Convert array.array to list
145                 lat = list(y) # Convert array.array to list
146                 fig.add_trace(go.Scattermapbox(
147                     lon=lon,
148                     lat=lat,
149                     mode='lines',
150                     line=dict(width=2, color='red'),
151                     hoverinfo='none',
152                     showlegend=False
153                 ))
154             elif geometry.type == 'LineString':
155                 x, y = geometry.xy
156                 lon = list(x) # Convert array.array to list
157                 lat = list(y) # Convert array.array to list
158                 fig.add_trace(go.Scattermapbox(
159                     lon=lon,
160                     lat=lat,
161                     mode='lines',
162                     line=dict(width=2, color='red'),
163                     hoverinfo='none',
164                     showlegend=False
165                 ))
166
167 # Step 3: Finalize and display the map
168 fig.update_layout(
169     title='Georgia Census Tracts Median Income and Top Wealth
170         Disparities',
171     mapbox_accesstoken=mapbox_access_token, # Ensure you have a valid
172         Mapbox access token
173     margin={'r': 0, 't': 60, 'l': 0, 'b': 0},
174     height=800
175 )
176 fig.show()
177 # =====
178 # analyzing the correlation
179
180 # Calculate average food insecurity rates for each tract pair
181 top_disparities_with_metrics['avg_pct_food_insecure'] = (
182     top_disparities_with_metrics['pct_food_insecure'] +
183     top_disparities_with_metrics['pct_food_insecure_neighbor']
184 ) / 2
185
186 # Ensure 'disparity' and 'avg_pct_food_insecure' are numeric
187 top_disparities_with_metrics = top_disparities_with_metrics.dropna(
188     subset=['disparity', 'avg_pct_food_insecure'])
189
190 # Pearson Correlation
191 pearson_corr = top_disparities_with_metrics['disparity'].corr(

```

```

191     top_disparities_with_metrics['avg_pct_food_insecure'],
192     method='pearson'
193 )
194 print(f"Pearson Correlation: {pearson_corr}")
195
196 # Spearman Correlation
197 spearman_corr = top_disparities_with_metrics['disparity'].corr(
198     top_disparities_with_metrics['avg_pct_food_insecure'],
199     method='spearman'
200 )
201 print(f"Spearman Correlation: {spearman_corr}")
202
203 # Extract food insecurity rates
204 low-fi = low_disparity_metrics['pct_food_insecure'].dropna()
205 high-fi = high_disparity_metrics['pct_food_insecure'].dropna()
206
207 # Perform t-test
208 t_stat, p_value = ttest_ind(low-fi, high-fi, equal_var=False)
209 print(f"T-Statistic: {t_stat}, P-Value: {p_value}")
210
211 # Interpret the results
212 if p_value < 0.05:
213     print("There is a statistically significant difference between
        high and low disparity tracts.")
214 else:
215     print("No statistically significant difference found.")

```

Website Code

This section outlines the structure of a GitHub Pages website for hosting and sharing the interactive choropleths. The site is organized into sections, each presenting maps and insights derived from the analysis.

Follow [this guide](#) for making a website with GitHub pages.

This website was built using the following directory structure:

Parent Directory

```

|-- Choropleths
|   |--FBCounties.html
|   |-- ...
|-- LICENSE
|-- README.md
|-- index.html

```

```

1  <!DOCTYPE html>
2  <html>
3      <head>
4
5      <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.0-beta3/dist
        /css/bootstrap.min.css" rel="stylesheet"

```



```

6      integrity="sha384-e0JMYsd53ii+sc0/bJGFsiCZc+5NDVN2yr8+0
      RDqr0Ql0h+rpP48ckxlpbzKgwra6" crossorigin="anonymous">
7  <title>Food Bank Group 1: Interactive Choropleths</title>
8  </head>
9  <body>
10     <div class="container">
11         <h1>Food Bank Group 1: Interactive Choropleths</h1>
12         <p>A collection of the Food Bank (Group 1)'s Interactive
            Choropleths.</p>
13         <small>Website and Choropleths by Annie Dmitrieff</small>
14
15         <hr>
16
17         <h2>Food Bank Service Regions</h2>
18         <p>Choropleths mapping Food Bank Service Regions across the
            state of Georgia.</p>
19
20         <h3>State-Wide Food Bank Service Regions</h3>
21         <iframe src="Choropleths/FBCounties.html" width="950" height=
            "800" style="border:1px solid black;"> </iframe>
22         <p>A map of the Food Bank providers' service regions across
            Georgia, grouped by county.</p>
23
24         <hr>
25
26         <h2>Food Insecurity Rates</h2>
27         <p>A collection of choropleths plotting Food-Insecurity rates
            by County and Census Tract for both the entire state of
            Georgia and the Food Bank of Northeast Georgia's service
            region.</p>
28
29         <h3>State-Wide Food Insecurity Rates by Tract</h3>
30         <p>A choropleth of the Food Insecurity rate by Census Tract
            across the state of Georgia. Data sourced from Mind the
            Meal Gap (2019-2022).</p>
31         <iframe src="Choropleths/FI_State_Tract.html" width="950"
            height="800" style="border:1px solid black;"> </iframe>
32
33         <h3>State-Wide Food Insecurity Rates by County</h3>
34         <p>A choropleth of the Food Insecurity rate by county across
            the state of Georgia. Data sourced from Mind the Meal Gap
            (2019-2022).</p>
35         <iframe src="Choropleths/FI_State_County.html" width="950"
            height="800" style="border:1px solid black;"> </iframe>
36
37         <h3>State-Wide Food Insecurity Rates by Tract Within Service
            Region</h3>
38         <p>A choropleth of the Food Insecurity rate by Census Tract
            within the Food Bank of Northeast Georgia's service
            region. Data sourced from Mind the Meal Gap (2019-2022).
            </p>
39         <iframe src="Choropleths/FIRate_FBNEGA_Tract.html" width="950
            " height="800" style="border:1px solid black;"> </iframe
            >

```

```

40
41 <h3>Wealth Disparity Borders with Census Tract Food
    Insecurity Rates</h3>
42 <p>A choropleth of the Food Insecurity Rates by Census Tract
    across the state, with Wealth Disparity boundaries
    between tracts plotted. Data sourced from Mind the Meal
    Gap (2019-2022).</p>
43 <iframe src="Choropleths/WealthDisp_FI.html" width="950"
    height="800" style="border:1px solid black;"> </iframe>
44
45 <hr>
46
47 <h2>Low-Access Metrics</h2>
48 <p>A collection of choropleths plotting Low-Access metrics by
    County and Census Tract for both the entire state of
    Georgia and the Food Bank of Northeast Georgia's service
    region.</p>
49
50 <h3>State-Wide Low-Access Tracts at 1, 10, and 20 Miles from
    a Food Source by Tract</h3>
51 <p>A choropleth of Low-Access Tracts at 1, 10, and 20 miles
    from a food source across the state of Georgia. Data
    sourced from the USDA Food Atlas (2022).</p>
52 <iframe src="Choropleths/
    LATractsOneTenTwenty_State_Tract.html" width="950"
    height="800" style="border:1px solid black;"> </iframe>
53
54 <h3>State-Wide Low-Access (1/2 Mile) Child Populations by
    County</h3>
55 <p>A choropleth showing the percentage of children (under 18)
    living more than a half-mile away from the nearest food
    source by county. Data sourced from the USDA Food Atlas
    (2022).</p>
56 <iframe src="Choropleths/LAKidsHalf_State_County.html" width=
    "950" height="800" style="border:1px solid black;">
    </iframe>
57
58 <h3>State-Wide Low-Access (1 Mile) Child Populations by
    County</h3>
59 <p>A choropleth showing the percentage of children (under 18)
    living more than one mile away from the nearest food
    source by county. Data sourced from the USDA Food Atlas
    (2022).</p>
60 <iframe src="Choropleths/LAKidsOne_State_County.html" width=
    "950" height="800" style="border:1px solid black;">
    </iframe>
61
62 <h3>State-Wide Low-Access (1/2 Mile) Child Populations by
    Tract</h3>
63 <p>A choropleth showing the percentage of children (under 18)
    living more than a half-mile away from the nearest food
    source by Census Tract. Data sourced from the USDA Food
    Atlas (2022).</p>
64 <iframe src="Choropleths/LAKidsHalf_State_Tract.html" width=

```

```

        "950" height="800" style="border:1px solid black; ">
    </iframe>
65
66    <h3>Low-Access (1/2 Mile) Child Populations by Tract Within
        Service Region</h3>
67    <p>A choropleth showing the percentage of children (under 18)
        living more than a half-mile away from the nearest food
        source by Census Tract within the Food Bank of Northeast
        Georgia's Service Region. Data sourced from the USDA Food
        Atlas (2022).</p>
68    <iframe src="Choropleths/LAKidsHalf_FBNEGA_Tract.html" width=
        "950" height="800" style="border:1px solid black; ">
        </iframe>
69
70    <h3>Low-Access (1 Mile) Child Populations by Tract Within
        Service Region</h3>
71    <p>A choropleth showing the percentage of children (under 18)
        living more than one mile away from the nearest food
        source by Census Tract within the Food Bank of Northeast
        Georgia's Service Region. Data sourced from the USDA Food
        Atlas (2022).</p>
72    <iframe src="Choropleths/LAKids1_FBNEGA_Tract.html" width=
        "950" height="800" style="border:1px solid black; ">
        </iframe>
73
74    <h3>State-Wide Low-Access (1/2 Mile) Senior Populations by
        County</h3>
75    <p>A choropleth showing the percentage of seniors (over 56)
        living more than a half-mile away from the nearest food
        source by county. Data sourced from the USDA Food Atlas
        (2022).</p>
76    <iframe src="Choropleths/LASnrsHalf_State_County" width="950"
        height="800" style="border:1px solid black; "> </iframe>
77
78    <h3>State-Wide Low-Access (1 Mile) Senior Populations by
        County</h3>
79    <p>A choropleth showing the percentage of seniors (over 56)
        living more than one mile away from the nearest food
        source by county. Data sourced from the USDA Food Atlas
        (2022).</p>
80    <iframe src="Choropleths/LASnrs1_State_County" width="950"
        height="800" style="border:1px solid black; "> </iframe>
81
82    <h3>State-Wide Low-Access (1/2 Mile) Senior Populations by
        Tract</h3>
83    <p>A choropleth showing the percentage of seniors (over 65)
        living more than a half-mile away from the nearest food
        source by Census Tract. Data sourced from the USDA Food
        Atlas (2022).</p>
84    <iframe src="Choropleths/LASnrssHalf_State_Tract.html" width=
        "950" height="800" style="border:1px solid black; ">
        </iframe>
85
86    <h3>State-Wide Low-Access (1 Mile) Senior Populations by

```

```

Tract</h3>
87 <p>A choropleth showing the percentage of seniors (over 65)
    living more than one mile away from the nearest food
    source by Census Tract. Data sourced from the USDA Food
    Atlas (2022).</p>
88 <iframe src="Choropleths/LASnrs1_State_Tract.html" width=
    "950" height="800" style="border:1px solid black;">
    </iframe>
89
90 <h3>Low-Access (1/2 Mile) Senior Populations by Tract Within
    Service Region</h3>
91 <p>A choropleth showing the percentage of seniors (over 65)
    living more than a half-mile away from the nearest food
    source by Census Tract within the Food Bank of Northeast
    Georgia's Service Region. Data sourced from the USDA Food
    Atlas (2022).</p>
92 <iframe src="Choropleths/LASnrsHalf_FBNEGA_Tract.html" width=
    "950" height="800" style="border:1px solid black;">
    </iframe>
93
94 <h3>Low-Access (1 Mile) Senior Populations by Tract Within
    Service Region</h3>
95 <p>A choropleth showing the percentage of seniors (over 65)
    living more than one mile away from the nearest food
    source by Census Tract within the Food Bank of Northeast
    Georgia's Service Region. Data sourced from the USDA Food
    Atlas (2022).</p>
96 <iframe src="Choropleths/LASnrs1_FBNEGA_Tract.html" width=
    "950" height="800" style="border:1px solid black;">
    </iframe>
97
98 <h3>Low-Access (1/2 Mile) Housing Units with No Vehicle by
    Tract Within Service Region</h3>
99 <p>A choropleth showing the percentage of housing units with
    no vehicle located more than a half-mile away from the
    nearest food source by Census Tract within the Food Bank
    of Northeast Georgia's Service Region. Data sourced from
    the USDA Food Atlas (2022).</p>
100 <iframe src="Choropleths/LAHalf_HUNV_FBNEGA_Tract.html"
    width="950" height="800" style="border:1px solid black;">
    </iframe>
101
102 <h3>Low-Access (1 Mile) Housing Units with No Vehicle by
    Tract Within Service Region</h3>
103 <p>A choropleth showing the percentage of housing units with
    no vehicle located more than a half-mile away from the
    nearest food source by Census Tract within the Food Bank
    of Northeast Georgia's Service Region. Data sourced from
    the USDA Food Atlas (2022).</p>
104 <iframe src="Choropleths/LA1_HUNV_FBNEGA_Tract.html" width=
    "950" height="800" style="border:1px solid black;">
    </iframe>
105
106 <h3>State-Wide Low-Access (1/2 Mile) SNAP-Enrolled Housing

```

```

107         Units by County</h3>
108     <p>A choropleth showing the percentage of SNAP-enrolled
        housing units located more than a half-mile away from the
        nearest food source by county. Data sourced from the
        USDA Food Atlas (2022).</p>
109     <iframe src="Choropleths/LASnapHalfCounty.html" width="950"
110         height="800" style="border:1px solid black;"> </iframe>
111
112     <h3>State-Wide Low-Access (1 Mile) SNAP-Enrolled Housing
        Units by County</h3>
113     <p>A choropleth showing the percentage of SNAP-enrolled
        housing units located more than one mile away from the
        nearest food source by county. Data sourced from the USDA
        Food Atlas (2022).</p>
114     <iframe src="Choropleths/LASnapHalfCounty.html" width="950"
115         height="800" style="border:1px solid black;"> </iframe>
116
117     <h3>Low-Access (1/2 Mile) SNAP-Enrolled Housing Units by
        Tract Within Service Region</h3>
118     <p>A choropleth showing the percentage of SNAP-enrolled
        housing units located more than a half-mile away from the
        nearest food source by Census Tract within the Food Bank
        of Northeast Georgia's Service Region. Data sourced from
        the USDA Food Atlas (2022).</p>
119     <iframe src="Choropleths/LAHalf_SNAPHU_FBNEGA_Tract.html"
120         width="950" height="800" style="border:1px solid black;">
        </iframe>
121
122     <h3>Low-Access (1 Mile) SNAP-Enrolled Housing Units by Tract
        Within Service Region</h3>
123     <p>A choropleth showing the percentage of SNAP-enrolled
        housing units located more than one mile away from the
        nearest food source by Census Tract within the Food Bank
        of Northeast Georgia's Service Region. Data sourced from
        the USDA Food Atlas (2022).</p>
124     <iframe src="Choropleths/LA1_SNAPHU_FBNEGA_Tract.html" width=
125         "950" height="800" style="border:1px solid black;">
        </iframe>
126
127     <hr>
128
129     <h2>Choropleths with Transportation Routes</h2>
130     <p>Choropleths of Low-Access and Food-Insecure metrics
        plotted with accessible public transportation routes
        within the service region.</p>
131
132     <h3>Public Transportation Routes Within the Food Bank of
        Northeast Georgia's Service Region</h3>
133     <p>A map of the counties within the Food Bank of Northeast
        Georgias service regions, as well as all public
        transportation routes within the entire service region.
        Data sourced from the U.S. Department of Transportation.
        </p>
134     <iframe src="Choropleths/transpaths_FBNEGA.html" width="950"

```

```

130         height="800" style="border:1px solid black;"> </iframe>
131     <h3>Public Transportation Routes with Food Insecurity Rates
        Within the Food Bank of Northeast Georgia's Service
        Region</h3>
132     <p>A choropleth of the Food Insecurity (FI) rate by Census
        Tract within the Food Bank of Northeast Georgias
        service region, overlayed by all public transportation
        routes available within the entire service region. Data
        sourced from Mind the Meal Gap (2019-2022) and the U.S.
        Department of Transportation.</p>
133     <iframe src="Choropleths/TRANS_FIRate_FBNEGA_Tract.html"
        width="950" height="800" style="border:1px solid black;">
        </iframe>
134
135     <h3>Public Transportation Routes with Low-Access (1/2 Mile)
        SNAP-Enrolled Housing Units by Tract Within Service
        Region</h3>
136     <p>A choropleth showing the percentage of SNAP-enrolled
        housing units located more than a half-mile away from the
        nearest food source by Census Tract within the Food Bank
        of Northeast Georgia's Service Region, overlayed with
        public transportation routes. Data sourced from the USDA
        Food Atlas (2022) and the U.S. Department of
        Transportation.</p>
137     <iframe src="Choropleths/
        TRANS_LAHalf_SNAPHU_FBNEGA_Tract.html" width="950"
        height="800" style="border:1px solid black;"> </iframe>
138
139     <h3>Public Transportation Routes with Low-Access (1 Mile)
        SNAP-Enrolled Housing Units by Tract Within Service
        Region</h3>
140     <p>A choropleth showing the percentage of SNAP-enrolled
        housing units located more than one mile away from the
        nearest food source by Census Tract within the Food Bank
        of Northeast Georgia's Service Region, overlayed with
        public transportation routes. Data sourced from the USDA
        Food Atlas (2022) and the U.S. Department of
        Transportation.</p>
141     <iframe src="Choropleths/TRANS_LA1_SNAPHU_FBNEGA_Tract.html"
        width="950" height="800" style="border:1px solid black;">
        </iframe>
142
143     <br>
144 </div>
145 </body>
146
147
148 </html>

```

Main.R

```
library(readxl)
library(dplyr)
library(ggplot2)
library(stringr)

source("~/GitHub/Food-Bank-Group-I/Data_Extraction.R", echo=TRUE)
source("~/GitHub/Food-Bank-Group-I/exploratory_analysis_food_bank.R",
echo=TRUE)

if (FALSE)
{
  MMG_data = extract_from_mmg_excel()
  MMG_data = MMG_data %>% mutate(a = round(`% of individuals in
households below 130% of poverty line that are food insecure` * `# of
individuals that are food insecure`),
                                b = round(`% of individuals in
households above 130% of poverty line that are food insecure` * `# of
individuals that are food insecure`),
                                c = round(`% of children in households
below 185% of poverty line that are food insecure` * `# of children that
are food insecure`),
                                d = round(`% of children in households
above 185% of poverty line that are food insecure` * `# of children that
are food insecure`))
  names(MMG_data) = c(names(MMG_data)[!names(MMG_data) %in% c("a", "b",
"c", "d")],
                      "# of individuals in households below 130% of
poverty line that are food insecure",
                      "# of individuals in households above 130% of
poverty line that are food insecure",
                      "# of children in households below 185% of poverty
line that are food insecure",
                      "# of children in households above 185% of poverty
line that are food insecure")
  MMG_data = transpose_data(MMG_data, "County", "State", "Label")

  SNAP_data = extract_from_SNAP_excel()

  DP02_data = extract_from_acs_csv("./data/DP02_csv", "Error|Percent")
  DP05_data = extract_from_acs_csv("./data/DP05_csv", "Error|Percent")
  S0101_data = extract_from_acs_csv("./data/S0101_csv",
"Error|Percent|Male|Female")
  S1501_data = extract_from_acs_csv("./data/S1501_csv",
"Error|Percent|Male|Female")
  S1701_data = extract_from_acs_csv("./data/S1701_csv",
"Error|Percent|Male|Female|poverty", 2012)
```



```

S1701_poverty_data = extract_from_acs_csv("./data/S1701_csv",
"Error|Percent|Male|Female|Total", 2012)
S2001_data = extract_from_acs_csv("./data/S2001_csv",
"Error|Male|Female|Percent")

S1701_data$`ALL INDIVIDUALS WITH INCOME BELOW THE FOLLOWING POVERTY
RATIOS` = rbind(S1701_data$`All Individuals below:`, S1701_data$`ALL
INDIVIDUALS WITH INCOME BELOW THE FOLLOWING POVERTY RATIOS`)
S1701_data$`All Individuals below:` = NULL
}

cor_factor = "SNAP All Persons Non-Public Assistance Participation"
#cor_factor = "Calc: SNAP Total PA and Non-PA People"
#cor_factor = "# of individuals that are food insecure"

if (FALSE)
{
  cor_averages = data.frame(Label = NULL, Correlation = NULL)
  for (i in 1:length(names))
  {
    print(names[i])

    exclude = c("# of children in households above 185% of poverty line
that are food insecure",
                "# of children in households below 185% of poverty line
that are food insecure",
                "# of children that are food insecure",
                "# of individuals in households above 130% of poverty
line that are food insecure",
                "# of individuals in households below 130% of poverty
line that are food insecure",
                "100 percent of poverty level",
                "200 percent of poverty level",
                "185 percent of poverty level",
                "150 percent of poverty level",
                "50 percent of poverty level")

    temp_data = get_significant_factors(names[i], cor_factor, exclude)

    mismatched = !(temp_data$Label %in% cor_averages$Label)

    for (n in 1:length(mismatched))
    {
      if (!mismatched[n])
      {
        cor_averages[cor_averages$Label == temp_data$Label[n], 2] =
as.numeric(cor_averages[cor_averages$Label == temp_data$Label[n], 2]) +
round(temp_data[n, 2])
      }
      else

```

```

    {
      cor_averages = rbind(cor_averages, temp_data[n,])
      cor_averages[cor_averages$Label == temp_data$Label[n], 2] =
round(cor_averages[cor_averages$Label == temp_data$Label[n], 2])
    }
  }

#cor_averages[, 2] = cor_averages[, 2] / length(names)

county_cor = NULL

for (i in 1:length(names))
{
  exclude = c("# of children in households above 185% of poverty line
that are food insecure",
              "# of children in households below 185% of poverty line
that are food insecure",
              "# of children that are food insecure",
              "# of individuals in households above 130% of poverty
line that are food insecure",
              "# of individuals in households below 130% of poverty
line that are food insecure")

  temp_data = get_factors(names[i], cor_factor, exclude)
  temp_data = temp_data %>% arrange(Label)

  if (is.null(county_cor))
  {
    county_cor = temp_data
  }
  else
  {
    county_cor = cbind(county_cor, temp_data[,2])
  }
}
names(county_cor) = c("Label", names)
}

county_data = list()
neg_prop_data = list()
pos_prop_data = list()

prop_data = NULL

for (i in 1:(length(names) + 1))
{
  print(names[i])
  if (!length(county_data) < 16)
  {

```

```

    temp_data = county_data[[i]]
  }
  if (i == 16)
  {
    temp_data = get_all()
  }
  else
  {
    temp_data = get_data(names[i])
  }
  county_data[[i]] = temp_data

#temp_data = county_data[[i]]

neg_data = data.frame(rep(names[i]),
                      temp_data$Year,
                      "% of population participating in SNAP",
                      "All",
                      temp_data$`Calc: SNAP Total PA and Non-PA
People` / temp_data$`Total population`)
neg_data = as.data.frame(mapply(c, neg_data,
                              data.frame(rep(names[i]),
                                          temp_data$Year,
                                          "% of population food
insecure",
                                          "All",
                                          temp_data$`# of individuals
that are food insecure` / temp_data$`Total population`)))
neg_data = as.data.frame(mapply(c, neg_data,
                              data.frame(rep(names[i]),
                                          temp_data$Year,
                                          "% of population over 65 years",
                                          "Negative",
                                          temp_data$`65 years and over` /
temp_data$`Total population`)))
neg_data = as.data.frame(mapply(c, neg_data,
                              data.frame(rep(names[i]),
                                          temp_data$Year,
                                          "% of population with a disability",
                                          "Negative",
                                          temp_data$`With a disability` /
temp_data$`Total population`)))
neg_data = as.data.frame(mapply(c, neg_data,
                              data.frame(rep(names[i]),
                                          temp_data$Year,
                                          "% of population divorced",
                                          "Negative",
                                          temp_data$`Divorced` / temp_data$`Total
population`)))
neg_data = as.data.frame(mapply(c, neg_data,

```

```

                                data.frame(rep(names[i]),
                                                temp_data$Year,
                                                "% of population in high
school",
                                "Negative",
                                temp_data$`High school
(grades 9-12)` / temp_data$`Total population`)))
    neg_data = as.data.frame(mapply(c, neg_data,
                                data.frame(rep(names[i]),
                                                temp_data$Year,
                                                "% of population over 25
with only a high school diploma",
                                "Negative",
                                temp_data$`High school
graduate (includes equivalency)` / temp_data$`Total population`)))
    names(neg_data) = c("County, State",
                        "Year",
                        "Label",
                        "Category",
                        "Percent")
    pos_data = data.frame(rep(names[i]),
                        temp_data$Year,
                        "% of population unemployed",
                        "Positive",
                        temp_data$`Unemployed` / temp_data$`Total
population`)
    pos_data = as.data.frame(mapply(c, pos_data,
                                data.frame(rep(names[i]),
                                                temp_data$Year,
                                                "% of population that are
veterans",
                                "Positive",
                                temp_data$`Civilian
veterans` / temp_data$`Total population`)))
    pos_data = as.data.frame(mapply(c, pos_data,
                                data.frame(rep(names[i]),
                                                temp_data$Year,
                                                "% of population in
kindergarten",
                                "Positive",
                                temp_data$`Kindergarten` /
temp_data$`Total population`)))
    pos_data = as.data.frame(mapply(c, pos_data,
                                data.frame(rep(names[i]),
                                                temp_data$Year,
                                                "% of population under 100
percent of poverty level",
                                "Positive",
                                temp_data$`100 percent of
poverty level` / temp_data$`Total population`)))

```

[illegible]

```

                                "Race / Ethnicity",
                                temp_data$`White` /
temp_data$`Total population`)))
  race_data = as.data.frame(mapply(c, race_data,
                                data.frame(rep(names[i]),
                                temp_data$Year,
                                "% of population that are
Native Hawaiian and Other Pacific Islander",
                                "Race / Ethnicity",
                                temp_data$`Native Hawaiian
and Other Pacific Islander` / temp_data$`Total population`)))
  race_data = as.data.frame(mapply(c, race_data,
                                data.frame(rep(names[i]),
                                temp_data$Year,
                                "% of population that are
some other race",
                                "Race / Ethnicity",
                                temp_data$`Some other
race` / temp_data$`Total population`)))
  names(race_data) = c("County, State",
                      "Year",
                      "Label",
                      "Category",
                      "Percent")

  prop_data = rbind(prop_data, neg_data)
  prop_data = rbind(prop_data, pos_data)
  prop_data = rbind(prop_data, race_data)
}

prop_data$Percent = as.numeric(prop_data$Percent)
prop_data$`County, State` = replace_na(prop_data$`County, State`,
"Service Region")

#prop_data = transpose_data(prop_data, "County, State", "Label")
names(county_data) = names

```

GeneratePlots.R

```

library(dplyr)
library(gganimate)
library(ggplot2)
library(tidyr)
library(sf)
library(gifski)
library(plotly)

if (FALSE)

```

```

{
  prop_data$Label = factor(prop_data$Label, unique(prop_data$Label))

  plot = prop_data %>%
    filter(Year == 2022
           & Category %in% c("Positive", "All")
           & Label != "% of population food insecure") %>%
    ggplot(aes(x = `County, State`, y = Percent, fill = Label)) +
    geom_bar(position = "dodge", stat = "identity") +
    scale_x_discrete(labels = c("Banks", "Barrow", "Clarke", "Elbert",
"Franklin", "Habersham", "Hart", "Jackson", "Madison", "Oconee",
"Oglethorpe", "Rabun", "Stephens", "Towns", "White", "Service Region"))
+
    #scale_y_continuous(limits = c(0, 0.4)) +
    facet_wrap(~Label, scale = "free") +
    theme(axis.text.x = element_text(angle = 270, vjust = 0, hjust=0),
legend.position = "none",
          axis.title = element_blank()) +
    ggtitle("SNAP Participation and Positively Correlated Factors,
2022")
  print(plot)

  plot = prop_data %>%
    filter(Year == 2022
           & Category %in% c("Negative", "All")
           & Label != "% of population food insecure") %>%
    ggplot(aes(x = `County, State`, y = Percent, fill = Label)) +
    geom_bar(position = "dodge", stat = "identity") +
    scale_x_discrete(labels = c("Banks", "Barrow", "Clarke", "Elbert",
"Franklin", "Habersham", "Hart", "Jackson", "Madison", "Oconee",
"Oglethorpe", "Rabun", "Stephens", "Towns", "White", "Service Region"))
+
    #scale_y_continuous(limits = c(0, 0.4)) +
    facet_wrap(~Label, scale = "free") +
    theme(axis.text.x = element_text(angle = 270, vjust = 0, hjust=0),
legend.position = "none",
          axis.title = element_blank()) +
    ggtitle("SNAP Participation and Negatively Correlated Factors,
2022")
  print(plot)

  plot = prop_data %>%
    filter(Year == 2022
           & Category %in% c("Race / Ethnicity", "All")
           & Label != "% of population food insecure") %>%
    ggplot(aes(x = `County, State`, y = Percent, fill = Label)) +
    geom_bar(position = "dodge", stat = "identity") +
    scale_x_discrete(labels = c("Banks", "Barrow", "Clarke", "Elbert",
"Franklin", "Habersham", "Hart", "Jackson", "Madison", "Oconee",

```



```

"Oglethorpe", "Rabun", "Stephens", "Towns", "White", "Service Region"))
+
  #scale_y_continuous(limits = c(0, 0.4)) +
  facet_wrap(~Label, scale = "free") +
  theme(axis.text.x = element_text(angle = 270, vjust = 0, hjust=0),
legend.position = "none",
        axis.title = element_blank()) +
  ggtitle("SNAP Participation and Race / Ethnicity Factors, 2022")
print(plot)
}

```

```

fixer = function(gp)
{
  lapply(1:length(gp$x$data), function(m)
  {
    gp$x$data[[m]]$hoveron <- "fills" # hover on the fill, not line
    if(length(gp$x$data[[m]]$text > 1)) {
      gp$x$data[[m]]$text <- gp$x$data[[m]]$text[1] # only one tooltip
per county
    }
  })
  gp
}

```

```

geo_data = read_sf("./data/georgia-counties.json")
geo_data = geo_data %>% filter(NAME %in% c("Banks",
                                           "Barrow",
                                           "Clarke",
                                           "Elbert",
                                           "Franklin",
                                           "Habersham",
                                           "Hart",
                                           "Jackson",
                                           "Madison",
                                           "Oconee",
                                           "Oglethorpe",
                                           "Rabun",
                                           "Stephens",
                                           "Towns",
                                           "White"))

```

```

if (FALSE)
{
  for (i in 2012:2022)
  {
    temp_data = prop_data %>%
      mutate(County = sub(",", ".", "", `County, State`)) %>%
      filter(Label == "Average meal cost in dollars" & !County ==
"Service Region")

```

```

temp_data = left_join(geo_data, temp_data, by = c("NAMELSAD" =
"County"))

temp_data = temp_data %>% filter(Year == i)

map = ggplot(temp_data) +
  geom_sf(aes(fill = Percent, geometry = geometry)) +
  geom_sf_text(aes(label = NAME), color = "white") +
  ggtitle(paste("Average Meal Cost by County,", i, sep = " ")) +
  theme_bw() +
  theme(axis.line = element_blank(),
        axis.ticks = element_blank(),
        axis.text = element_blank(),
        axis.title = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        panel.border = element_blank(),
        panel.background = element_blank()) +
  scale_fill_viridis_c(
    limits = c(2.5, 4.5),
    breaks = c(2.5, 3, 3.5, 4, 4.5),
    name = "Meal Cost in Dollars",
    guide = guide_legend(
      label.position = "bottom",
      title.position = "top",
      nrow = 1
    )
  )
  )
  ggsave(paste(paste("./images/meal_cost_choropleth_", i, sep = " "),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
}
}

if (FALSE)
{
  temp_data = prop_data %>%
    mutate(County = sub(".*", "", `County, State`)) %>%
    mutate(Tooltip = paste(paste(paste("Average meal cost in ", County),
": "), Percent)) %>%
    filter(Label == "Average meal cost in dollars" & !County == "Service
Region")

  temp_data = left_join(geo_data, temp_data, by = c("NAMELSAD" =
"County"))

  map = ggplot(temp_data) +
    geom_sf(aes(fill = Percent, geometry = geometry, frame = Year, group
= NAME)) +
    #geom_sf_text(aes(label = NAME), color = "white") +
    labs(title = 'Average Meal Cost by County, 2012-2022') +

```

```

theme_bw() +
theme(axis.line = element_blank(),
      axis.ticks = element_blank(),
      axis.text = element_blank(),
      axis.title = element_blank(),
      panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(),
      panel.border = element_blank(),
      panel.background = element_blank()) +
scale_fill_viridis_c(
  limits = c(2.5, 4.5),
  breaks = c(2.5, 3, 3.5, 4, 4.5),
  name = "Meal Cost in Dollars"
) #+
#transition_manual(as.numeric(Year))
#animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
ggplotly(map) %>%
  animation_opts(transition = 0) %>% fixer()
}

generate_choropleth = function(gео, data, label, title, legend, year =
2022)
{
  temp_data = data %>%
    mutate(County = sub(".*", "", `County, State`)) %>%
    filter(Label == label & !County == "Service Region")

  temp_data = left_join(gео, temp_data, by = c("NAMELSAD" = "County"))

  i = year

  temp_data = temp_data %>% filter(Year == i)

  map = ggplot(temp_data) +
    geom_sf(aes(fill = Percent, geometry = geometry)) +
    geom_sf_text(aes(label = NAME), color = "white") +
    ggtitle(paste(title, i, sep = ", ")) +
    theme_bw() +
    theme(axis.line = element_blank(),
          axis.ticks = element_blank(),
          axis.text = element_blank(),
          axis.title = element_blank(),
          panel.grid.major = element_blank(),
          panel.grid.minor = element_blank(),
          panel.border = element_blank(),
          panel.background = element_blank()) +
    scale_fill_viridis_c(
      name = legend,

```

```

    limits = c(min(temp_data$Percent) - 0.01, max(temp_data$Percent) +
0.01),
    breaks = c(round(min(temp_data$Percent), 2),
                round((max(temp_data$Percent) - min(temp_data$Percent))
/ 4 + min(temp_data$Percent), 2),
                round((max(temp_data$Percent) - min(temp_data$Percent))
* 2 / 4 + min(temp_data$Percent), 2),
                round((max(temp_data$Percent) - min(temp_data$Percent))
* 3 / 4 + min(temp_data$Percent), 2),
                round(max(temp_data$Percent), 2)),
    guide = guide_legend(
      label.position = "bottom",
      title.position = "top",
      nrow = 1
    )
  )
}

return(map)
}

```

```

generate_animated_choropleth = function(gео, data, label, title, legend)
{
  temp_data = data %>%
    mutate(County = sub(",", ".", "", `County, State`)) %>%
    filter(Label == label & !County == "Service Region")

  temp_data = left_join(gео, temp_data, by = c("NAMELSAD" = "County"))

  map = ggplot(temp_data) +
    geom_sf(aes(fill = Percent, geometry = geometry)) +
    geom_sf_text(aes(label = NAME), color = "white") +
    ggtitle(paste(title, "2012 to 2022", sep = ", ")) +
    theme_bw() +
    theme(axis.line = element_blank(),
          axis.ticks = element_blank(),
          axis.text = element_blank(),
          axis.title = element_blank(),
          panel.grid.major = element_blank(),
          panel.grid.minor = element_blank(),
          panel.border = element_blank(),
          panel.background = element_blank()) +
    scale_fill_viridis_c(
      name = legend,
      limits = c(min(temp_data$Percent) - 0.01, max(temp_data$Percent) +
0.01),
      breaks = c(round(min(temp_data$Percent), 2),
                  round((max(temp_data$Percent) - min(temp_data$Percent))
/ 4 + min(temp_data$Percent), 2),
                  round((max(temp_data$Percent) - min(temp_data$Percent))
* 2 / 4 + min(temp_data$Percent), 2),

```

```

        round((max(temp_data$Percent) - min(temp_data$Percent))
* 3 / 4 + min(temp_data$Percent), 2),
        round(max(temp_data$Percent), 2)),
    guide = guide_legend(
      label.position = "bottom",
      title.position = "top",
      nrow = 1
    )
  ) +
  transition_manual(as.numeric(Year))

```

```

  return(map)
}

```

```
i = 2022
```

```

if (FALSE)
{
  map = generate_choropleth(geo_data, prop_data, "% of population
unemployed", "Unemployment by County", "% of population")
  ggsave(paste(paste("./images/unemployment_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
  map = generate_choropleth(geo_data, prop_data, "% of population that
are veterans", "Veterans by County", "% of population")
  ggsave(paste(paste("./images/veterans_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
  map = generate_choropleth(geo_data, prop_data, "% of population in
kindergarten", "Kindergarteners by County", "% of population")
  ggsave(paste(paste("./images/kindergarten_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
  map = generate_choropleth(geo_data, prop_data, "% of population under
100 percent of poverty level", "Poverty Rate by County", "% of
population")
  ggsave(paste(paste("./images/poverty_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
  map = generate_choropleth(geo_data, prop_data, "% of population in
college or graduate school", "College / Graduate Students by County", "%
of population")
  ggsave(paste(paste("./images/college_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)

  map = generate_choropleth(geo_data, prop_data, "% of population over
65 years", "Seniors by County", "% of population")
  ggsave(paste(paste("./images/seniors_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
  map = generate_choropleth(geo_data, prop_data, "% of population with a
disability", "Disability by County", "% of population")
  ggsave(paste(paste("./images/disability_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)

```

```

map = generate_choropleth(geo_data, prop_data, "% of population
divorced", "Individuals Currently Divorced by County", "% of
population")
ggsave(paste(paste("./images/divorce_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
map = generate_choropleth(geo_data, prop_data, "% of population in
high school", "High Schoolers by County", "% of population")
ggsave(paste(paste("./images/highschool_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
map = generate_choropleth(geo_data, prop_data, "% of population over
25 with only a high school diploma", "Individuals Over 25 With Only a
High School Diploma by County", "% of population")
ggsave(paste(paste("./images/diploma_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)

map = generate_choropleth(geo_data, prop_data, "Average meal cost in
dollars", "Average Meal Cost by County", "Meal cost in dollars")
ggsave(paste(paste("./images/meal_choropleth_", i, sep = ""), ".png",
sep = ""), map, units = "in", width = 9.6, height = 5.4)
map = generate_choropleth(geo_data, prop_data, "Median earnings in
dollars", "Median Earnings by County", "Median income in dollars")
ggsave(paste(paste("./images/income_choropleth_", i, sep = ""),
".png", sep = ""), map, units = "in", width = 9.6, height = 5.4)
map = generate_choropleth(geo_data, prop_data, "% of population
participating in SNAP", "SNAP Participation by County", "% of
population")
ggsave(paste(paste("./images/snap_choropleth_", i, sep = ""), ".png",
sep = ""), map, units = "in", width = 9.6, height = 5.4)
}

```

```

map = generate_animated_choropleth(geo_data, prop_data, "% of population
unemployed", "Unemployment by County", "% of population")
animate(map, renderer = gifschi_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/unemployed_animated.gif", animation =
last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
that are veterans", "Veterans by County", "% of population")
animate(map, renderer = gifschi_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/veterans_animated.gif", animation =
last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
in kindergarten", "Kindergarteners by County", "% of population")
animate(map, renderer = gifschi_renderer(), units = "px", width = 640,
height = 360)

```

```

anim_save("./images/kindergarten_animated.gif", animation =
last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
under 100 percent of poverty level", "Poverty Rate by County", "% of
population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/poverty_animated.gif", animation = last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
in college or graduate school", "College / Graduate Students by County",
"% of population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/college_animated.gif", animation = last_animation())

map = generate_animated_choropleth(geo_data, prop_data, "% of population
over 65 years", "Seniors by County", "% of population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/seniors_animated.gif", animation = last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
with a disability", "Disability by County", "% of population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/disability_animated.gif", animation =
last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
divorced", "Individuals Currently Divorced by County", "% of
population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/divorced_animated.gif", animation =
last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
in high school", "High Schoolers by County", "% of population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/high_school_animated.gif", animation =
last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
over 25 with only a high school diploma", "Individuals Over 25 With Only
a High School Diploma by County", "% of population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)

map = generate_animated_choropleth(geo_data, prop_data, "Average meal
cost in dollars", "Average Meal Cost by County", "Meal cost in dollars")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/meal_animated.gif", animation = last_animation())

```

```

map = generate_animated_choropleth(geo_data, prop_data, "Median earnings
in dollars", "Median Earnings by County", "Median income in dollars")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/income_animated.gif", animation = last_animation())
map = generate_animated_choropleth(geo_data, prop_data, "% of population
participating in SNAP", "SNAP Participation by County", "% of
population")
animate(map, renderer = gifski_renderer(), units = "px", width = 640,
height = 360)
anim_save("./images/snap_animated.gif", animation = last_animation())

```

DataExtraction.R

```

library(readxl)
library(dplyr)
library(ggplot2)
library(tidyr)

names = c("Banks County, Georgia",
          "Barrow County, Georgia",
          "Clarke County, Georgia",
          "Elbert County, Georgia",
          "Franklin County, Georgia",
          "Habersham County, Georgia",
          "Hart County, Georgia",
          "Jackson County, Georgia",
          "Madison County, Georgia",
          "Oconee County, Georgia",
          "Oglethorpe County, Georgia",
          "Rabun County, Georgia",
          "Stephens County, Georgia",
          "Towns County, Georgia",
          "White County, Georgia")

extract_from_acs_csv = function(folder_path, exclude, start_year = 2010)
{
  folder = list.files(folder_path, pattern = "^ACS", full.names = TRUE)
  data = list()

  for (n in 1:length(folder))
  {
    temp_data = read.csv(folder[n])
    temp_data = temp_data[ , !grepl(exclude, names(temp_data))]
    temp_data$Year = start_year + n - 1
    temp_data[,1] = gsub("\u00A0", "", temp_data[,1])
    temp_data = data.frame(lapply(temp_data, function(x) gsub("[,]", "",
x)))
    names(temp_data) = c("Label", names, "Year")
  }
}

```



```

temp_data = temp_data[c("Label", "Year", names)]

header_indices = which(temp_data[,3] == "")
headers = temp_data[header_indices, 1]
headers = gsub("\\s*\\([^\\)]+\\)", "", headers)
headers = gsub("\\u00A0", "", headers)

header_indices = c(header_indices, nrow(temp_data) + 1)

if (header_indices[1] != 1)
{
  header_indices = c(1, header_indices)
  headers = c("FIRST", headers)
}

for (i in 1:length(headers))
{
  start = header_indices[i] + 1
  end = header_indices[i + 1] - 1

  if (length(data) == 0 | !all(headers[i] %in% names(data)))
  {
    data[[headers[i]]] = data.frame(matrix(ncol = length(names) + 2,
nrow = 0))
  }

  data[[headers[i]]] = rbind(data[[headers[i]]],
temp_data[start:end,])

  row.names(data[[i]]) = NULL
}
}

return(data)
}

transpose_data = function(data, id_col, new_id_col, group_col = "Year")
{
  transposed_data = NULL

  region = !names(data) %in% c(id_col, group_col)
  groups = unlist(unique(data[group_col]), use.names = FALSE)

  for (i in 1:length(groups))
  {
    temp_data = data %>% filter(data[group_col] == groups[i])
    temp_data = temp_data[region]
    temp_data = t(temp_data)
    temp_data = data.frame(temp_data)
  }
}

```

```

    temp_data = temp_data %>% mutate(colnames(data)[region],
rep(groups[i]))

    colnames(temp_data) = c(unlist(unique(data[id_col])), use.names =
FALSE), new_id_col, group_col)
    rownames(temp_data) = NULL

    temp_data = temp_data[c(new_id_col, group_col,
unlist(unique(data[id_col])), use.names = FALSE)]

    transposed_data = rbind(transposed_data, temp_data)
  }

  return(transposed_data)
}

remove_duplicates = function(data, id_col = "Label", group_col = "Year")
{
  groups = unlist(unique(data[group_col]), use.names = FALSE)
  return_data = NULL

  for (i in 1:length(groups))
  {
    temp_data = data %>% filter(data[group_col] == groups[i])
    temp_data = temp_data[!duplicated(temp_data[id_col]),]
    return_data = rbind(return_data, temp_data)
  }

  return(return_data)
}

sum_duplicates = function(data, id_col = "Label", group_col = "Year")
{
  groups = unlist(unique(data[group_col]), use.names = FALSE)
  return_data = NULL

  for (i in 1:length(groups))
  {
    temp_data = data %>% filter(data[group_col] == groups[i])

    first = duplicated(temp_data$Label, fromLast = TRUE)
    last = duplicated(temp_data$Label)
    unique = !first & !last

    temp_data[first, !colnames(temp_data) %in% c(id_col, group_col)] =
data.frame(lapply(temp_data[first, !colnames(temp_data) %in% c(id_col,
group_col)], function(x) as.numeric(x))) +
data.frame(lapply(temp_data[last, !colnames(temp_data) %in% c(id_col,
group_col)], function(x) as.numeric(x)))
    temp_data = temp_data[first | unique,]
  }
}

```

```

    return_data = rbind(return_data, temp_data)
  }

  return(return_data)
}

extract_from_mmg_excel = function()
{
  categories = c("County, State",
                 "Year",
                 "% of total population that are food insecure",
                 "# of individuals that are food insecure",
                 "% of individuals in households below 130% of poverty
line that are food insecure",
                 "% of individuals in households above 130% of poverty
line that are food insecure",
                 "% of children that are food insecure",
                 "# of children that are food insecure",
                 "% of children in households below 185% of poverty line
that are food insecure",
                 "% of children in households above 185% of poverty line
that are food insecure",
                 "Average meal cost",
                 "Weighted annual food budget shortfall")

  MMG_folder = list.files("./data/MMG_Data", pattern = "^MMG",
full.names = TRUE)
  MMG_data = NULL

  temp_sheet = read_excel(MMG_folder[1], sheet = "County")
  temp_data = temp_sheet %>% filter(`State Name` == "GA"
                                & `County Code` %in% c("BANKS",
"BARROW",
"CLARKE",
"ELBERT",
"FRANKLIN",
"HABERSHAM",
"HART",
"JACKSON",
"MADISON",
"OCONEE",
"OGLETHORPE",
"RABUN",
"STEPHENS",
"TOWNS",
"WHITE"))

  temp_data$Year = rep(2009)
  temp_data[11] = temp_data[11] + temp_data[12]
  temp_data[2] = names
  temp_data = temp_data[,c(2, 19, 4, 5, 10, 11, 15:18, 14, 13)]

```

```

names(temp_data) = categories

MMG_data = temp_data

for (i in 0:7)
{
  sheet_name = excel_sheets(MMG_folder[i + 2])
  sheet_name = grep("County", sheet_name, value = TRUE)
  temp_sheet = read_excel(MMG_folder[i + 2], sheet = sheet_name[1])
  temp_data = temp_sheet %>% filter(`County, State` %in% names)
  temp_data$Year = rep(2010 + i)
  temp_data[11] = as.numeric(unlist(temp_data[11], use.names = FALSE))
  temp_data[12] = as.numeric(unlist(temp_data[11], use.names = FALSE))
  temp_data[11] = temp_data[11] + temp_data[12]
  temp_data = temp_data[,c(3, 19, 4, 5, 10, 11, 13:18)]
  names(temp_data) = categories
  temp_data[3] = as.numeric(unlist(temp_data[3], use.names = FALSE))
  temp_data[5] = as.numeric(unlist(temp_data[5], use.names = FALSE))
  temp_data[7] = as.numeric(unlist(temp_data[7], use.names = FALSE))
  temp_data[9] = as.numeric(unlist(temp_data[9], use.names = FALSE))
  temp_data[10] = as.numeric(unlist(temp_data[10], use.names = FALSE))

  MMG_data = MMG_data %>% rbind(temp_data)
}

temp_sheet = read_excel(MMG_folder[10], sheet = "2018 County", skip =
1)
temp_data = temp_sheet %>% filter(`County, State` %in% names)

temp_data$Year = rep(2018)
temp_data[11] = as.numeric(unlist(temp_data[11], use.names = FALSE))
temp_data[12] = as.numeric(unlist(temp_data[11], use.names = FALSE))
temp_data[11] = temp_data[11] + temp_data[12]
temp_data[2] = names
temp_data = temp_data[,c(3, 19, 4, 5, 10, 11, 13:18)]
names(temp_data) = categories
temp_data[3] = as.numeric(unlist(temp_data[3], use.names = FALSE))
temp_data[5] = as.numeric(unlist(temp_data[5], use.names = FALSE))
temp_data[7] = as.numeric(unlist(temp_data[7], use.names = FALSE))
temp_data[9] = as.numeric(unlist(temp_data[9], use.names = FALSE))
temp_data[10] = as.numeric(unlist(temp_data[10], use.names = FALSE))

MMG_data = MMG_data %>% rbind(temp_data)

# 2019 to 2022
temp_sheet = read_excel(MMG_folder[11], sheet = "County")
temp_data = temp_sheet %>% filter(`County, State` %in% names)

temp_data = temp_data[,c(3, 4, 5, 6, 11:17, 19)]

```

```

temp_data = temp_data %>% arrange(Year)
names(temp_data) = categories
temp_data[3] = as.numeric(unlist(temp_data[3], use.names = FALSE))
temp_data[5] = as.numeric(unlist(temp_data[5], use.names = FALSE))
temp_data[7] = as.numeric(unlist(temp_data[7], use.names = FALSE))
temp_data[9] = as.numeric(unlist(temp_data[9], use.names = FALSE))
temp_data[10] = as.numeric(unlist(temp_data[10], use.names = FALSE))
temp_data[11] = as.numeric(unlist(temp_data[11], use.names = FALSE))
temp_data[12] = as.numeric(unlist(temp_data[12], use.names = FALSE))

MMG_data = MMG_data %>% rbind(temp_data)

return(MMG_data)
}

extract_from_SNAP_excel = function()
{
  SNAP_folder = list.files("./data/SNAP_Participation", pattern = "^J",
full.names = TRUE)
  SNAP_data = NULL

  for (i in 0:12)
  {
    temp_sheet = read_excel(SNAP_folder[i + 1], skip = 3)
    str = str_split_fixed(temp_sheet$`Substate/Region`, " ", n =
7)[,1:4]
    temp_sheet[c("FIPS", "State", "EBT", "County")] = str
    temp_data = temp_sheet[c("State",
                             "County",
                             "SNAP All Persons Public Assistance
Participation",
                             "SNAP All Persons Non-Public Assistance
Participation",
                             "Calc: SNAP Total PA and Non-PA People")]
    temp_data = temp_data %>%
      filter(State == "GA"
             & County %in% c("BANKS",
                              "BARROW",
                              "CLARKE",
                              "ELBERT",
                              "FRANKLIN",
                              "HABERSHAM",
                              "HART",
                              "JACKSON",
                              "MADISON",
                              "OCONEE",
                              "OGLETHORPE",
                              "RABUN",
                              "STEPHENS",
                              "TOWNS",

```

```

        "WHITE"))

    pop = data.frame(Year = rep(2010 + i), t(temp_data[,3:5]))
    colnames(pop) = c("Year", names)

    SNAP_data = SNAP_data %>% rbind(pop)
  }

  SNAP_data = SNAP_data %>% mutate(Label = rep(c("SNAP All Persons
Public Assistance Participation",
                                                "SNAP All Persons Non-
Public Assistance Participation",
                                                "Calc: SNAP Total PA
and Non-PA People"), 13))
  SNAP_data = SNAP_data[c("Label", "Year", names)]

  rownames(SNAP_data) = NULL

  return(SNAP_data)
}

get_mismatched_names = function(data, id_col = "Label", group_col =
"Year")
{
  groups = unlist(unique(data[group_col]), use.names = FALSE)

  occurrences = table(data["Label"])

  return(occurrences[occurrences < length(groups)])
}

exclude_percentages = function(data)
{
  cols = names(data)

  temp_data = data

  for (col in 1:length(cols))
  {
    if (all(temp_data[cols[col]] == 0, na.rm = TRUE))
    {
      next
    }
    else if (all(temp_data[cols[col]] <= 1, na.rm = TRUE))
    {
      temp_data[cols[col]] = NULL
    }
  }
}

return(temp_data)

```

```

}

convert_percentages = function(data, total_pop)
{
  return_data = NULL
  for (i in 1:nrow(data))
  {
    if (any(grepl("%", data[i,])))
    {
      pop = total_pop %>% filter(Year == data[i, "Year"])
      pop = pop[,-c(1, 2)]
      pop = as.numeric(pop)

      temp_data = gsub("%", "", data[i,-c(1, 2)])
      temp_data = as.numeric(temp_data) / 100

      temp_data = round(temp_data * pop)

      temp_data = c(data[i, 1], data[i, 2], temp_data)
    }
    else
    {
      temp_data = data[i,]
    }

    return_data = rbind(return_data, temp_data)
  }

  return(return_data)
}

```

Analysis.R

```

library(readxl)
library(dplyr)
library(ggplot2)

names = c("Banks County, Georgia",
          "Barrow County, Georgia",
          "Clarke County, Georgia",
          "Elbert County, Georgia",
          "Franklin County, Georgia",
          "Habersham County, Georgia",
          "Hart County, Georgia",
          "Jackson County, Georgia",
          "Madison County, Georgia",
          "Oconee County, Georgia",
          "Oglethorpe County, Georgia",
          "Rabun County, Georgia",

```

```
"Stephens County, Georgia",  
"Towns County, Georgia",  
"White County, Georgia")
```

```
format_analysis = function(data, county_name)  
{  
  data = transpose_data(data, "Label", "County, State")  
  data = data %>% filter(`County, State` == county_name)  
  data$Year = as.numeric(data$Year)  
  data = data[-1]  
}  
  
sum_all_analysis = function(data)  
{  
  return_data = NULL  
  for (i in 1:length(unique(data$Year)))  
  {  
    year_data = data %>% filter(Year == as.numeric(Year[1]) + i - 1)  
    rownames(year_data) = NULL  
    temp_data = year_data[, 1:3]  
    temp_data[, 3] =  
rowSums(data.frame(lapply(year_data[, 3:ncol(year_data)], as.numeric)))  
  
    return_data = rbind(return_data, temp_data)  
  }  
  return_data = transpose_data(return_data, "Label", "County, State")  
  return_data$Year = as.numeric(return_data$Year)  
  return_data = return_data[-1]  
}  
  
get_all = function()  
{  
  sex_and_age = remove_duplicates(DP05_data$`SEX AND AGE`)  
  sex_and_age = sex_and_age %>% filter(Label != "16 years and over")  
  sex_and_age = sex_and_age %>% filter(Label != "Sex ratio (males per  
100 females)")  
  sex_and_age = sex_and_age %>% filter(Label != "Under 18 years")  
  sex_and_age = sum_all_analysis(sex_and_age)  
  
  race = remove_duplicates(DP05_data$`Race alone or in combination with  
one or more other races`)  
  race$Label = gsub("Some Other Race", "Some other race", race$Label)  
  race = race %>% filter(Label != "Total population")  
  race = sum_all_analysis(race)  
  
  education = S1701_data$`EDUCATIONAL ATTAINMENT`  
  education = sum_all_analysis(education)  
  
  poverty = remove_duplicates(S1701_data$`ALL INDIVIDUALS WITH INCOME  
BELOW THE FOLLOWING POVERTY RATIOS`)
```



```

poverty = rbind(poverty, S1701_poverty_data$FIRST %>% filter(Label ==
"Population for whom poverty status is determined"))
poverty$Label = gsub("Population for whom poverty status is
determined", "100 percent of poverty level", poverty$Label)
poverty = poverty %>% filter(!Label %in%
names(get_mismatched_names(poverty)))
poverty = poverty %>% filter(!Label %in% c("Male", "Female"))
poverty = sum_all_analysis(poverty)

employment = remove_duplicates(S1701_data$`EMPLOYMENT STATUS`)
employment = employment %>% filter(!Label %in% c("Male", "Female"))
employment = sum_all_analysis(employment)

marriage = sum_duplicates(DP02_data$`MARITAL STATUS`)
marriage = sum_all_analysis(marriage)

school = DP02_data$`SCHOOL ENROLLMENT`
school = sum_all_analysis(school)

veteran = DP02_data$`VETERAN STATUS`
veteran = veteran %>% filter(Label == "Civilian veterans")
veteran = sum_all_analysis(veteran)

disability = remove_duplicates(DP02_data$`DISABILITY STATUS OF THE
CIVILIAN NONINSTITUTIONALIZED POPULATION`)
disability = disability %>% filter(disability[names[1]] != "(X)")
disability = disability %>% filter(!Label %in% c("Under 18 years", "18
to 64 years", "65 years and over"))
disability = sum_all_analysis(disability)

income = S2001_data$FIRST
income = remove_duplicates(income)
income$Label = gsub("FULL-TIME YEAR-ROUND WORKERS WITH EARNINGS",
"Full-time year-round workers with earnings", income$Label)
income = income %>% filter(!Label %in% c("Mean earnings (dollars)",
"Mean earnings (dollars) for
full-time year-round workers with earnings",
"Median earnings (dollars)
for full-time year-round workers with earnings"))

income = convert_percentages(income, income[income$Label == "Full-time
year-round workers with earnings",])

income = sum_all_analysis(income)

mmg = MMG_data %>% filter(!(grepl("%", MMG_data$Label) & !grepl("#",
MMG_data$Label)))
mmg = sum_all_analysis(mmg)
snap = sum_all_analysis(SNAP_data)

```

```

temp_data = inner_join(sex_and_age, race, by = "Year")
temp_data = inner_join(temp_data, education, by = "Year")
temp_data = inner_join(temp_data, poverty, by = "Year")
temp_data = inner_join(temp_data, employment, by = "Year")
temp_data = inner_join(temp_data, marriage, by = "Year")
temp_data = inner_join(temp_data, school, by = "Year")
temp_data = inner_join(temp_data, veteran, by = "Year")
temp_data = inner_join(temp_data, disability, by = "Year")
temp_data = inner_join(temp_data, income, by = "Year")
temp_data = inner_join(temp_data, mmg, by = "Year")
temp_data = inner_join(temp_data, snap, by = "Year")

combined_data = data.frame(apply(temp_data, 2, function(x)
as.numeric(as.character(x))))
names(combined_data) = names(temp_data)

return(combined_data)
}

get_data = function(county_name)
{
  sex_and_age = remove_duplicates(DP05_data$`SEX AND AGE`)
  sex_and_age = sex_and_age %>% filter(Label != "16 years and over")
  sex_and_age = sex_and_age %>% filter(Label != "Sex ratio (males per
100 females)")
  sex_and_age = sex_and_age %>% filter(Label != "Under 18 years")
  sex_and_age = format_analysis(sex_and_age, county_name)

  race = remove_duplicates(DP05_data$`Race alone or in combination with
one or more other races`)
  race$Label = gsub("Some Other Race", "Some other race", race$Label)
  race = race %>% filter(Label != "Total population")
  race = format_analysis(race, county_name)

  education = S1701_data$`EDUCATIONAL ATTAINMENT`
  education = format_analysis(education, county_name)

  poverty = remove_duplicates(S1701_data$`ALL INDIVIDUALS WITH INCOME
BELOW THE FOLLOWING POVERTY RATIOS`)
  poverty = rbind(poverty, S1701_poverty_data$FIRST %>% filter(Label ==
"Population for whom poverty status is determined"))
  poverty$Label = gsub("Population for whom poverty status is
determined", "100 percent of poverty level", poverty$Label)
  poverty = poverty %>% filter(!Label %in%
names(get_mismatched_names(poverty)))
  poverty = poverty %>% filter(!Label %in% c("Male", "Female"))
  poverty = format_analysis(poverty, county_name)

  employment = remove_duplicates(S1701_data$`EMPLOYMENT STATUS`)
  employment = employment %>% filter(!Label %in% c("Male", "Female"))

```

```

employment = format_analysis(employment, county_name)

marriage = sum_duplicates(DP02_data$`MARITAL STATUS`)
marriage = format_analysis(marriage, county_name)

school = DP02_data$`SCHOOL ENROLLMENT`
school = format_analysis(school, county_name)

veteran = DP02_data$`VETERAN STATUS`
veteran = veteran %>% filter(Label == "Civilian veterans")
veteran = format_analysis(veteran, county_name)

disability = remove_duplicates(DP02_data$`DISABILITY STATUS OF THE
CIVILIAN NONINSTITUTIONALIZED POPULATION`)
disability = disability %>% filter(disability[county_name] != "(X)")
disability = disability %>% filter(!Label %in% c("Under 18 years", "18
to 64 years", "65 years and over"))

disability = format_analysis(disability, county_name)

income = S2001_data$FIRST
income = remove_duplicates(income)
income$Label = gsub("FULL-TIME YEAR-ROUND WORKERS WITH EARNINGS",
"Full-time year-round workers with earnings", income$Label)
income = income %>% filter(!Label %in% c("Mean earnings (dollars)",
"Mean earnings (dollars) for
full-time year-round workers with earnings",
"Median earnings (dollars)
for full-time year-round workers with earnings"))

income = convert_percentages(income, income[income$Label == "Full-time
year-round workers with earnings",])

income = format_analysis(income, county_name)

mmg = format_analysis(MMG_data, county_name)
snap = format_analysis(SNAP_data, county_name)

temp_data = inner_join(sex_and_age, race, by = "Year")
temp_data = inner_join(temp_data, education, by = "Year")
temp_data = inner_join(temp_data, poverty, by = "Year")
temp_data = inner_join(temp_data, employment, by = "Year")
temp_data = inner_join(temp_data, marriage, by = "Year")
temp_data = inner_join(temp_data, school, by = "Year")
temp_data = inner_join(temp_data, veteran, by = "Year")
temp_data = inner_join(temp_data, disability, by = "Year")
temp_data = inner_join(temp_data, income, by = "Year")
temp_data = inner_join(temp_data, mmg, by = "Year")
temp_data = inner_join(temp_data, snap, by = "Year")

```

```

    combined_data = data.frame(apply(temp_data, 2, function(x)
as.numeric(as.character(x))))
    names(combined_data) = names(temp_data)

    return(combined_data)
}

get_significant_factors = function(county_name, factor_col, exclude =
c())
{
    county = get_data(county_name)
    county = exclude_percentages(county)

    county = county[,!colnames(county) %in% exclude]

    cor_matrix = cor(county)
    cor = data.frame(Label = rownames(cor_matrix), Correlation =
cor_matrix[,factor_col])
    sig = cor %>% filter(abs(Correlation) > 0.5)
    pos = sig %>% filter(Correlation > 0) %>% arrange(desc(Correlation))
    pos = pos[-1,]
    pos = head(pos, n = 5)
    neg = sig %>% filter(Correlation < 0) %>% arrange(Correlation)
    neg = head(neg, n = 5)

    complete = rbind(pos, neg)
    rownames(complete) = NULL
    complete = complete %>% arrange(desc(Correlation))

    return(complete)
}

get_factors = function(county_name, factor_col, exclude = c())
{
    county = get_data(county_name)
    county = exclude_percentages(county)

    county = county[,!colnames(county) %in% exclude]

    cor_matrix = cor(county)
    cor = data.frame(Label = rownames(cor_matrix), Correlation =
cor_matrix[,factor_col])
    cor = cor %>% arrange(desc(Correlation))
    cor = cor[-1,]
    rownames(cor) = NULL
    return(cor)
}

```

Using R:

```
#install.packages("readxl")  
library(readxl)
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A138:A143")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "B138:B143")
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#install.packages("ggplot2")  
library(ggplot2)
```

```
#reading first few rows of data  
head(foodbank)
```

```
#covert x to factor  
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of food insecure children in service region from 2018-2022
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Number of Food Insecure Children in Service Region from 2018-2022", x = "Year", y = "Number  
of Food Insecure Children")
```

```
#creating line graph of food insecure children in service region from 2018-2022
```

```
ggplot(foodbank, aes(x = x, y = y, group = 1)) + geom_line(color = "blue", size = 1) +  
geom_point(color = "red", size = 1) + labs(title = "Number of Food Insecure Children in Service  
Region from 2018-2022", x = "Year", y = "Number of Food Insecure Children")
```

```
#install.packages("tidyr")  
library(tidyr)
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A132:A137")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "B132:P137")
```

```
foodbank <- data.frame( x = x_data[[1]], y_data)
```

```
foodbank_long <- foodbank %>%  
  pivot_longer(cols = -x, names_to = "County", values_to = "y")
```

```
#creating line graph of food insecure children by county over time
```

```
ggplot(foodbank_long, aes(x=x, y=y, color = County, group = County)) + geom_line(size = 1) +  
  geom_point(size = 1) +  
  labs(title = "Number of Food Insecure Children by County from 2018-2022", x = "Year", y =  
  "Number of Food Insecure Children")
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A626:A631")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "B626:P631")
```

```
foodbank <- data.frame( x = x_data[[1]], y_data)
```

```
foodbank_long <- foodbank %>%  
  pivot_longer(cols = -x, names_to = "County", values_to = "y")
```

```
#creating line graph for child food insecurity rates by county over time
```

```
ggplot(foodbank_long, aes(x=x, y=y, color = County, group = County)) + geom_line(size = 1) +  
  geom_point(size = 1) +  
  labs(title = "Child Food Insecurity Rates by County between 2018-2022", x = "Year", y = "Child  
  Food Insecurity Rates")
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A160:A174")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "A548:A563")
```

```
head(x_data)  
head(y_data)
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#reading first few rows of data
```

```
head(foodbank)
```

```
#covert x to factor
```

```
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of food insecure children in service region in 2022
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Number of Food Insecure Children by County in 2022", x = "County", y = "Number of Food  
Insecure Children") + coord_flip()
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A160:A174")
```

```
y_data <- read_excel("Documents/foodbank.xlsx", range = "G160:G174")
```

```
head(x_data)
```

```
head(y_data)
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#reading first few rows of data
```

```
head(foodbank)
```

```
#covert x to factor
```

```
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of proportion of food insecure children
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Proportion of Food Insecure Children by School District in 2022 (Number of Food Insecure  
Children/ Total Number of Students Enrolled)", x = "County", y = "Proportion of Food Insecure  
Children") + coord_flip()
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A258:A272")
```

```
y_data <- read_excel("Documents/foodbank.xlsx", range = "B258:B272")
```

```
head(x_data)
```

```
head(y_data)
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#reading first few rows of data  
head(foodbank)
```

```
#covert x to factor  
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of funding in school districts
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Dollar Amount Spent per Student per Year by School District in 2022", x = "County", y = "Dollar  
Amount") + coord_flip()
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A491:A505")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "G491:G505")
```

```
head(x_data)  
head(y_data)
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#reading first few rows of data  
head(foodbank)
```

```
#covert x to factor  
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of food insecure seniors
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Number of Seniors Below Poverty Level by County in 2022", x = "County", y = "Number of  
Seniors (65+)") + coord_flip()
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "A491:A505")
```



```

y_data <- read_excel("Documents/foodbank.xlsx", range = "H491:H505")

head(x_data)
head(y_data)

foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])

#reading first few rows of data
head(foodbank)

#covert x to factor
foodbank$x <- as.factor(foodbank$x)

#creating bar graph of veterans in counties

ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =
"Number of Veterans by County in 2022", x = "County", y = "Number of Veterans") +
coord_flip()


#downloading excel sheet

x_data <- read_excel("Documents/foodbank.xlsx", range = "S492:S506")
y_data <- read_excel("Documents/foodbank.xlsx", range = "T492:T506")

head(x_data)
head(y_data)

foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])

#reading first few rows of data
head(foodbank)

#covert x to factor
foodbank$x <- as.factor(foodbank$x)

#creating bar graph of number of seniors more than 1 mile from supermarket

ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =
"Number of Seniors More than 1 Mile from a Supermarket by County in 2022", x = "County", y =
"Number of Seniors") + coord_flip()

```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "S492:S506")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "U492:U506")
```

```
head(x_data)  
head(y_data)
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#reading first few rows of data  
head(foodbank)
```

```
#covert x to factor  
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of number of children more than 1 mile from supermarket
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Number of Children More than 1 Mile from a Supermarket by County in 2022", x = "County", y  
= "Number of Children") + coord_flip()
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "S492:S506")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "V492:V506")
```

```
head(x_data)  
head(y_data)
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#reading first few rows of data  
head(foodbank)
```

```
#covert x to factor  
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of households with no vehicle more than 1 mile from supermarket
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Number of Housing Units without Vehicle Beyond 1 Mile from a Supermarket by County", x =  
"County", y = "Number of Households") + coord_flip()
```

```
#downloading excel sheet
```

```
x_data <- read_excel("Documents/foodbank.xlsx", range = "S492:S506")  
y_data <- read_excel("Documents/foodbank.xlsx", range = "W492:W506")
```

```
head(x_data)  
head(y_data)
```

```
foodbank <- data.frame( x = x_data[[1]], y = y_data[[1]])
```

```
#reading first few rows of data  
head(foodbank)
```

```
#covert x to factor  
foodbank$x <- as.factor(foodbank$x)
```

```
#creating bar graph of households receiving SNAP more than 1 mile from supermarket
```

```
ggplot(foodbank, aes(x = x, y = y,)) + geom_bar(stat = "identity", fill = "blue") + labs(title =  
"Number of Housing Units Receiving SNAP Benefits Beyond 1 Mile from a Supermarket by  
County", x = "County", y = "Number of Households") + coord_flip()
```

Food Bank R Code

Population Density

```
library(readxl)
Food_Bank_MMG_Data <- read_excel("Downloads/Food Bank MMG Data.xlsx",
+   sheet = "Population Density")
> View(Food_Bank_MMG_Data)
ggplot(Food_Bank_MMG_Data, aes(x = County, y = `2019`, fill = County)) +
+   geom_bar(stat = "identity") +
+   geom_text(aes(label = `2019`), vjust = -0.3, size = 3) +
+   labs(
+     title = "Population Density 2019",
+     x = "County",
+     y = "People per Square Mile"
+   ) +
+   theme_minimal() +
+   theme(
+     axis.text.x = element_text(angle = 45, hjust = 1),
+     legend.position = "none"
+   )
ggplot(Food_Bank_MMG_Data, aes(x = County, y = `2020`, fill = County)) +
+   geom_bar(stat = "identity") +
+   geom_text(aes(label = `2020`), vjust = -0.3, size = 3) +
+   labs(
+     title = "Population Density 2020",
+     x = "County",
+     y = "People per Square Mile"
+   ) +
+   theme_minimal() +
+   theme(
+     axis.text.x = element_text(angle = 45, hjust = 1),
+     legend.position = "none"
+   )
ggplot(Food_Bank_MMG_Data, aes(x = County, y = `2021`, fill = County)) +
+   geom_bar(stat = "identity") +
+   geom_text(aes(label = `2021`), vjust = -0.3, size = 3) +
+   labs(
+     title = "Population Density 2021",
+     x = "County",
+     y = "People per Square Mile"
+   ) +
+   theme_minimal() +
+   theme(
```

```

+   axis.text.x = element_text(angle = 45, hjust = 1),
+   legend.position = "none"
+ )
ggplot(Food_Bank_MMG_Data, aes(x = County, y = `2022`, fill = County)) +
+   geom_bar(stat = "identity") +
+   geom_text(aes(label = `2022`), vjust = -0.3, size = 3) +
+   labs(
+     title = "Population Density 2022",
+     x = "County",
+     y = "People per Square Mile"
+   ) +
+   theme_minimal() +
+   theme(
+     axis.text.x = element_text(angle = 45, hjust = 1),
+     legend.position = "none"
+   )

```

Food Insecurity Rate

```

library(readxl)
> Food_Bank_MMG_Data <- read_excel("Downloads/Food Bank MMG Data.xlsx",
+   sheet = "Food Insecurity Rate")
> View(Food_Bank_MMG_Data)
> install.packages(tidyverse)
> library(tidyverse)
> library(ggplot2)
#renaming the heading of the table
library(dplyr)
> # Rename the column 'County'
> Food_Bank_MMG_Data <- Food_Bank_MMG_Data %>%
+   rename(County = ...1)

#converting table from wide format to long
> library(tidyr)
> Food_Bank_MMG_Data_long <- gather(Food_Bank_MMG_Data, key = "Year", value =
"Rate", -County)
> # Plotting the line graph
> ggplot(Food_Bank_MMG_Data_long, aes(x = Year, y = Rate, color = County, group =
County)) +
+   geom_line() +
+   labs(title = "Food Insecurity Rate by County (2018-2022)",
+     x = "Year",
+     y = "Rate") +
+   theme_minimal()

```

Overall Trend

```
library(dplyr)
> library(ggplot2)
>
> # Convert data from wide format to long format
> library(tidyr)
> Food_Bank_MMG_Data_long <- gather(Food_Bank_MMG_Data, key = "Year", value =
"Rate", -County)
>
> # Calculate the average rate for each year
> average_rate_per_year <- Food_Bank_MMG_Data_long %>%
+   group_by(Year) %>%
+   summarise(Average_Rate = mean(Rate, na.rm = TRUE))
>
> # Plot the overall trend
> ggplot(average_rate_per_year, aes(x = as.numeric(Year), y = Average_Rate)) +
+   geom_line(color = "blue") +
+   geom_point(color = "red") +
+   labs(title = "Overall Trend of Food Insecurity Rate by Year",
+        x = "Year",
+        y = "Average Rate of Food Insecurity for all Counties") +
+   theme_minimal()
```

Unemployment Rate

```
library(readxl)
> Food_Bank_MMG_Data <- read_excel("Downloads/Food Bank MMG Data.xlsx",
+   sheet = "Unemployment Rate")
> View(Food_Bank_MMG_Data)
library(ggplot2)
>
> # Convert the data from wide format to long format
> library(tidyr)
> Food_Bank_MMG_Data_long <- gather(Food_Bank_MMG_Data, key = "Year", value =
"Rate", -County)
>
> # Plotting the line graph
> ggplot(Food_Bank_MMG_Data_long, aes(x = Year, y = Rate, color = County, group =
County)) +
+   geom_line() +
+   labs(title = "Unemployment Rate by County (2018-2022)",
+        x = "Year",
+        y = "Rate") +
+   theme_minimal()
```

Overall Trend

```
library(dplyr)
> library(ggplot2)
>
> # Convert data from wide format to long format
> library(tidyr)
> Food_Bank_MMG_Data_long <- gather(Food_Bank_MMG_Data, key = "Year", value =
"Rate", -County)
>
> # Calculate the average rate for each year
> average_rate_per_year <- Food_Bank_MMG_Data_long %>%
+   group_by(Year) %>%
+   summarise(Average_Rate = mean(Rate, na.rm = TRUE))
>
> # Plot the overall trend
> ggplot(average_rate_per_year, aes(x = as.numeric(Year), y = Average_Rate)) +
+   geom_line(color = "blue") +
+   geom_point(color = "red") +
+   labs(title = "Overall Trend of Unemployment Rate by Year",
+        x = "Year",
+        y = "Average Rate of Unemployment for all Counties") +
+   theme_minimal()
>
```

Median Income

```
library(readxl)
> Food_Bank_MMG_Data <- read_excel("Downloads/Food Bank MMG Data.xlsx",
+   sheet = "Median Income")
New names:
• `` -> `...1`
> View(Food_Bank_MMG_Data)
library(dplyr)
> # Rename the column 'County'
> Food_Bank_MMG_Data <- Food_Bank_MMG_Data %>%
+   rename(County = ...1)
library(ggplot2)
>
> # Convert the data from wide format to long format for easier plotting
> library(tidyr)
> Food_Bank_MMG_Data_long <- gather(Food_Bank_MMG_Data, key = "Year", value =
"Rate", -County)
>
> # Plotting the line graph
```

```

> ggplot(Food_Bank_MMG_Data_long, aes(x = Year, y = Rate, color = County, group =
County)) +
+   geom_line() +
+   labs(title = "Median Income by County (2019-2022)",
+         x = "Year",
+         y = "Median Income") +
+   theme_minimal()
>

> library(dplyr)
> library(ggplot2)
>
> # Convert data from wide format to long format
> library(tidyr)
> Food_Bank_MMG_Data_long <- gather(Food_Bank_MMG_Data, key = "Year", value =
"Rate", -County)
>
> # Calculate the average rate for each year
> average_rate_per_year <- Food_Bank_MMG_Data_long %>%
+   group_by(Year) %>%
+   summarise(Average_Rate = mean(Rate, na.rm = TRUE))
>
> # Plot the overall trend
> ggplot(average_rate_per_year, aes(x = as.numeric(Year), y = Average_Rate)) +
+   geom_line(color = "blue") +
+   geom_point(color = "red") +
+   labs(title = "Overall Trend of Median Income (2019-2022)",
+         x = "Year",
+         y = "Average Median Income for all Counties") +
+   theme_minimal()
>

```

SNAP Participation

```

library(readxl)
> Food_Bank_MMG_Data <- read_excel("Downloads/Food Bank MMG Data.xlsx",
+   sheet = "Snap Participation")
New names:
• `` -> `...1`
> # Rename the column 'County'
> Food_Bank_MMG_Data <- Food_Bank_MMG_Data %>%
+   rename(County=...1)

library(ggplot2)
>

```



```
> # Convert the data from wide format to long format for easier plotting
> library(tidyr)
> Food_Bank_MMG_Data_long <- gather(Food_Bank_MMG_Data, key = "Year", value =
"Rate", -County)
>
> # Plotting the line graph
> ggplot(Food_Bank_MMG_Data_long, aes(x = Year, y = Rate, color = County, group =
County)) +
+   geom_line() +
+   labs(title = "SNAP Participation Rate by County (2018-2022)",
+     x = "Year",
+     y = "Rate of Participation") +
+   theme_minimal()
>
```